

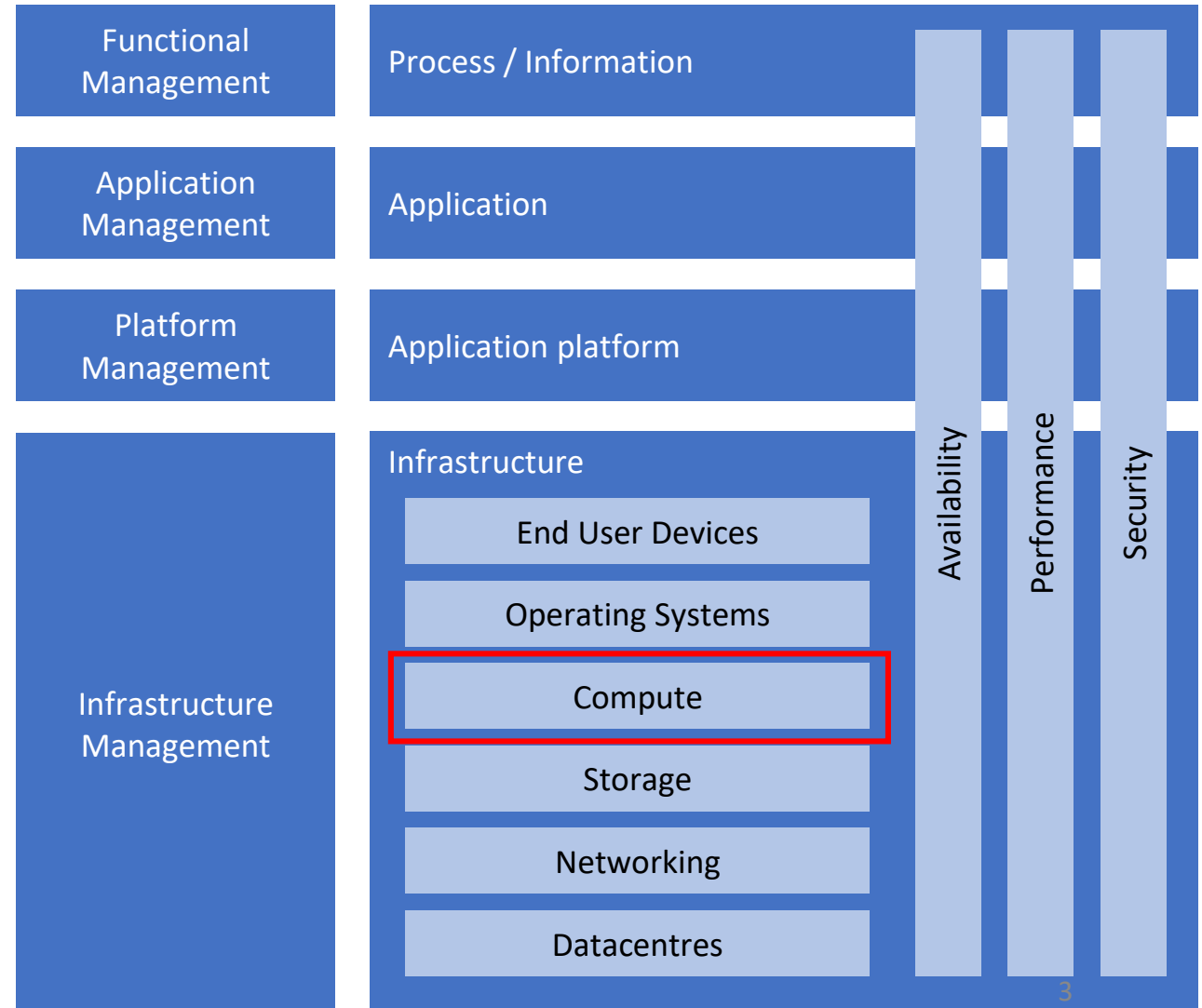
Chapter 6: Compute

Objectives

- Discuss compute building blocks
- Discuss compute availability
- Discuss compute performance
- Discuss computer security

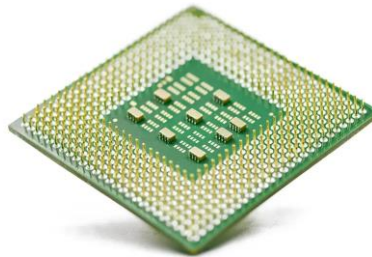
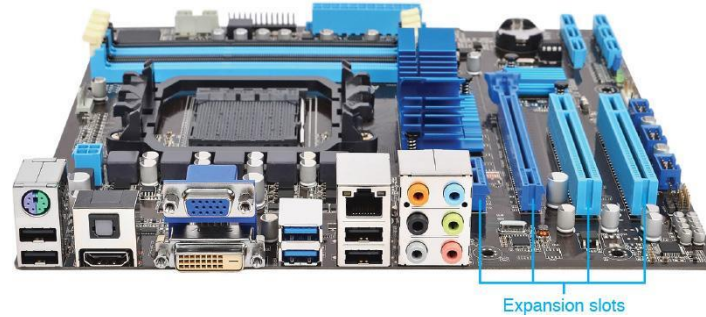
Introduction

- Compute refers to computers in a datacentre, both physical or virtual machines
- Three groups:
 - Mainframes
 - Midrange systems
 - x86 servers

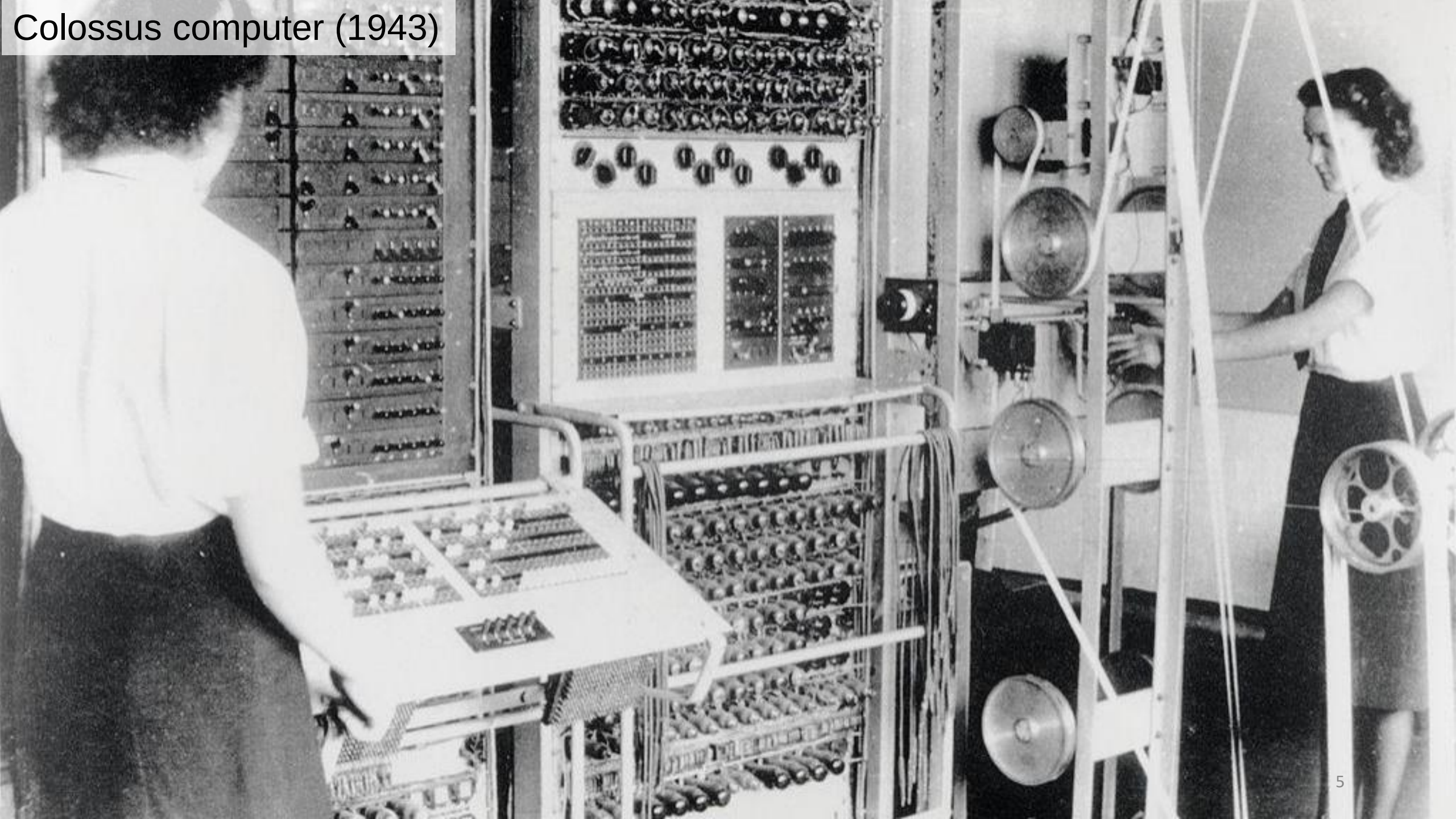


Introduction

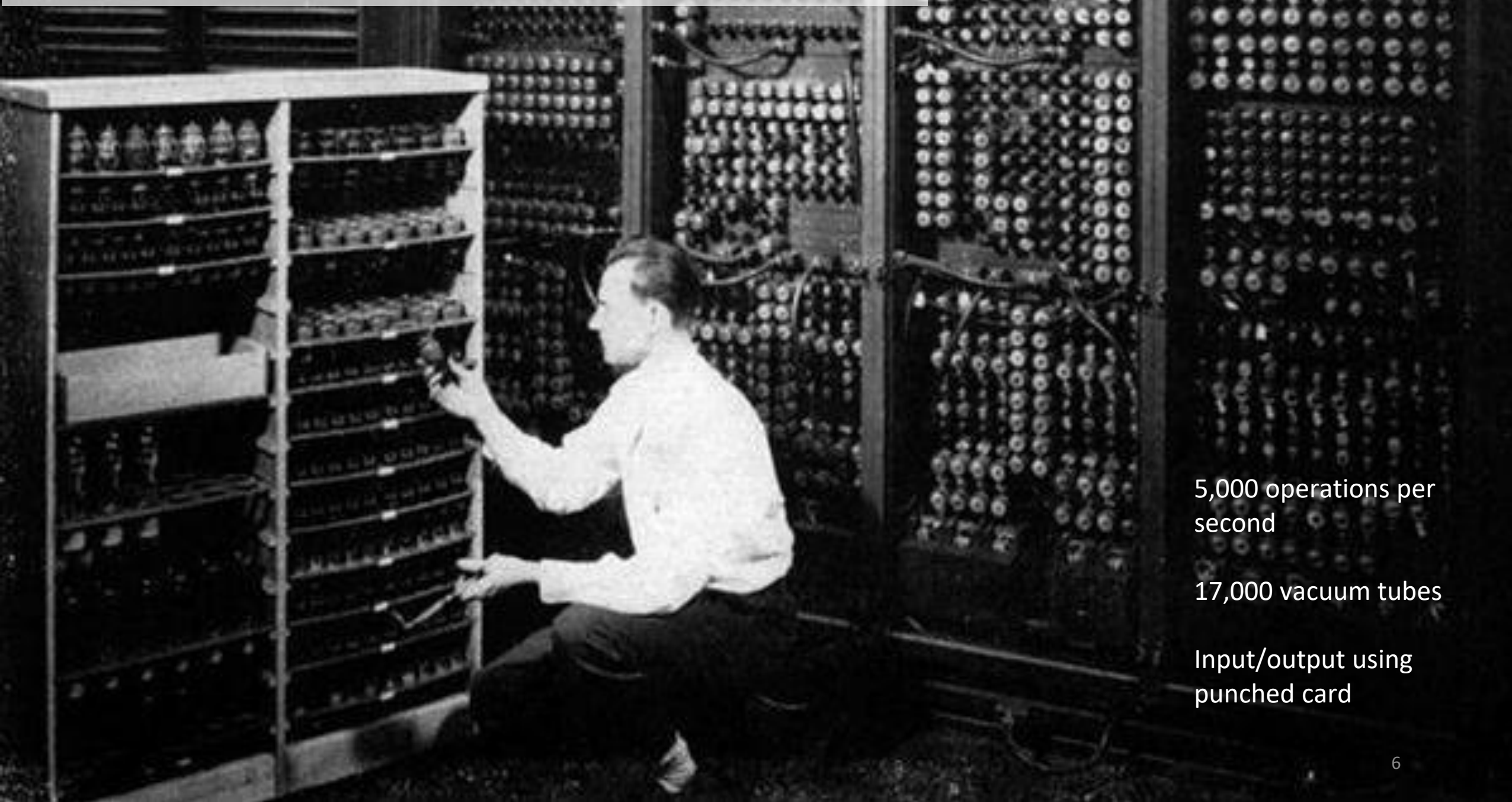
- Physical computers contain:



Colossus computer (1943)



Electronic Numerical Integrator And Computer (1945)



5,000 operations per second

17,000 vacuum tubes

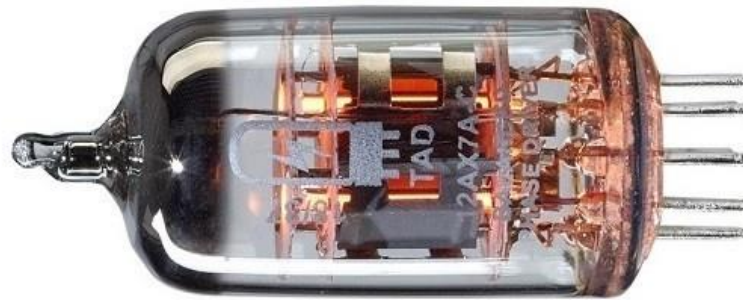
Input/output using punched card

THE IMITATION GAME



History

- In the 1960s computers started to be built using transistors instead of vacuum tubes
 - Smaller
 - Faster
 - Cheaper to produce
 - Required less power
 - Much more reliable



Vacuum
Tube



Transisto
r

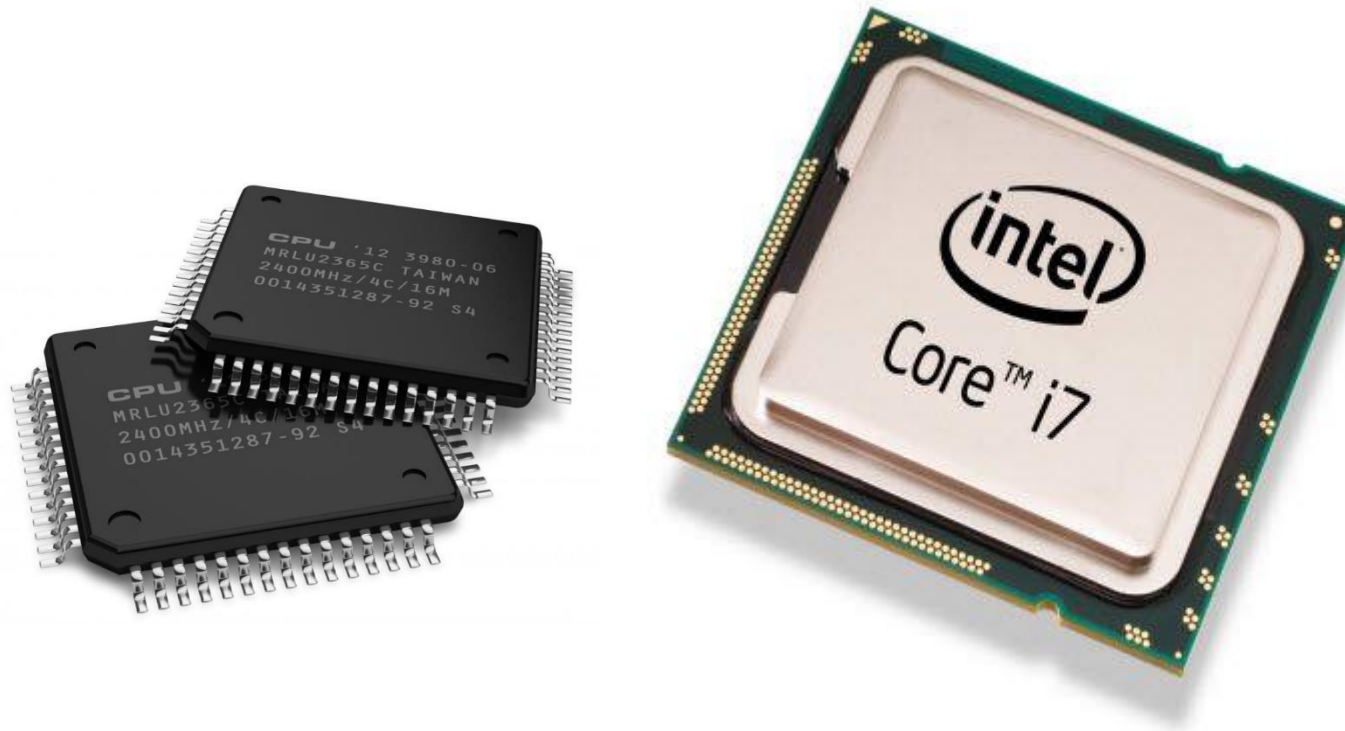
History

- The transistor based computers were followed in the 1970s by computers based on integrated circuit (IC) technology



History

- Microprocessors decreased size and cost of computers even further



Compute building blocks

Compute building blocks



Computer Housing



Processor



GPU



Memory

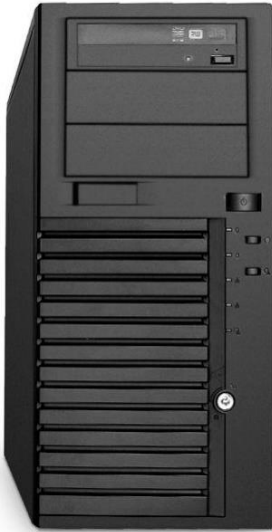


BIOS



Interfaces

Computer housing



Tower



Rack



Blade

Computer housing - Blade



+



HDD/SSD



Controller



SAN Switch

Processors

- The brain of a computer.
- It processes instructions and performs calculations.



Processors - speed

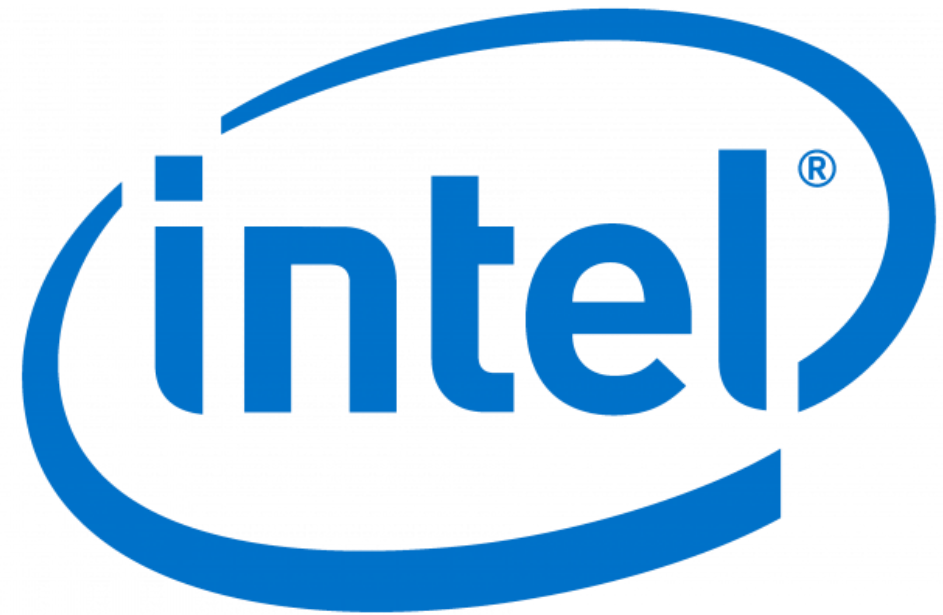
- A CPU needs a high frequency clock to operate, generating so-called clock ticks or clock cycles
 - Each machine code instruction takes one or more clock ticks to execute
 - An ADD instruction typically costs 1 tick to compute
- The speed at which the CPU operates is defined in GHz (billions of clock ticks per second)
 - A single core of a 2.4 GHz CPU can perform 2.4 billion additions in 1 second

Processors – word size

- Each CPU is designed to handle data in chunks, called words, with a specific size. Today, most CPUs have a word size of 64 bits
- The word size is reflected in many aspects of a CPU's structure and operation:
 - The majority of the internal memory registers are the size of one word
 - The largest piece of data that can be transferred to and from the working memory in a single operation is a word
 - A 64-bit CPU can address 17,179,869,184 TB of memory (64-bit word)

Intel x86 processors

- Intel CPUs became the de-facto standard for many computer architectures
- Since these names all ended with the number 86, the generic architecture was referred to as x86
- The latest Intel x86 model is the 24-core i9-14900K Processor, running on 6.0 GHz





INDUSTRY LEADING OPTIMIZED SILICON



AMD x86 processors

- Advanced Micro Devices, Inc. (AMD) is the second-largest global supplier of microprocessors based on the x86 architecture
- The latest model is the 16-core Ryzen 9950X CPU, running on 5.7 GHz



ARM processors

- The ARM (Advanced RISC Machine) is the most used CPU in the world
- In 2013, 10 billion ARM processors were shipped, running on:
- The CPU is produced by a large number of manufacturers under license of ARM
- Since 2016, ARM is owned by Japanese telecommunications company SoftBank Group

The ARM logo, consisting of the lowercase letters 'arm' in a bold, blue, sans-serif font.



CPUs created by computer manufacturers

- To be less dependent on the large CPU manufacturers, large computer manufacturers started creating their own CPU designs
- Public cloud providers also develop their own processors now
 - AWS developed the Graviton processors to deliver the best performance for cloud workloads. Graviton is a 64-bit ARM-based CPU. Today the most powerful CPU contain 64 cores, running at 2.4 GHz.
 - Microsoft also launched their own CPU: the Ampere Altra for their Azure cloud platform. It has up to 64 CPU cores and is also based on the ARM architecture
 - Google's Argos is a video optimized CPU for YouTube video encoding purposes







CPUs created by computer manufacturers

- In 2024, Qualcomm announced the Qualcomm Snapdragon X Plus and Snapdragon X Elite chips.
- Snapdragon chips use an ARM-based architecture rather than the x86 architecture.
- One problem with previous ARM-based laptops has been many popular programs simply aren't designed for it, instead being coded for x86 architectures. This requires either different low-level code or a translation layer.
- Microsoft has an emulator called Prism to run nonnative apps.



Up to

10-Core CPU

3.8 TFLOPS GPU[†]

45 TOPS NPU^{††}



Up to

12-Core CPU

3.8 TFLOPS GPU[†]

45 TOPS NPU^{††}

GPUs

- Graphics processing units (GPUs) can be used together with CPUs to accelerate specific calculations
- A GPU has a massively parallel architecture consisting of thousands of smaller, more efficient cores designed for handling multiple tasks simultaneously
- GPUs are used in Artificial Intelligence (AI) and Machine Learning (ML)
- GPUs are also used in game consoles

GPU as a Service Market

The global GPU as a service market is home to players like DigitalOcean, Amazon Web Services (AWS), Oracle Cloud, Microsoft Azure, OVHcloud, Google Cloud, Qarnot Computing, IBM Cloud, Packet, NVIDIA Cloud, Vultr, Scaleway, Igneous, HPE GreenLake, Paperspace, NetApp, Alibaba Cloud, Hetzner, Rescale, and Linode.

Regional Analysis

North America | Europe | Asia Pacific
Latin America | Middle East and Africa

2023-2030

CAGR
28.78%

By Delivery Model

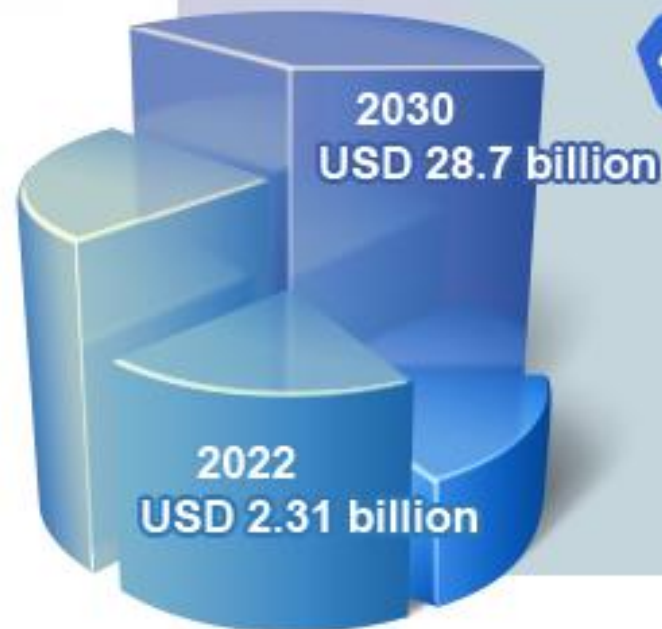
- Private Cloud
- Public Cloud
- Hybrid Cloud

By Component

- Services
- Software

By Application

- Real Estate
- Gaming
- Automotive
- Healthcare
- Others



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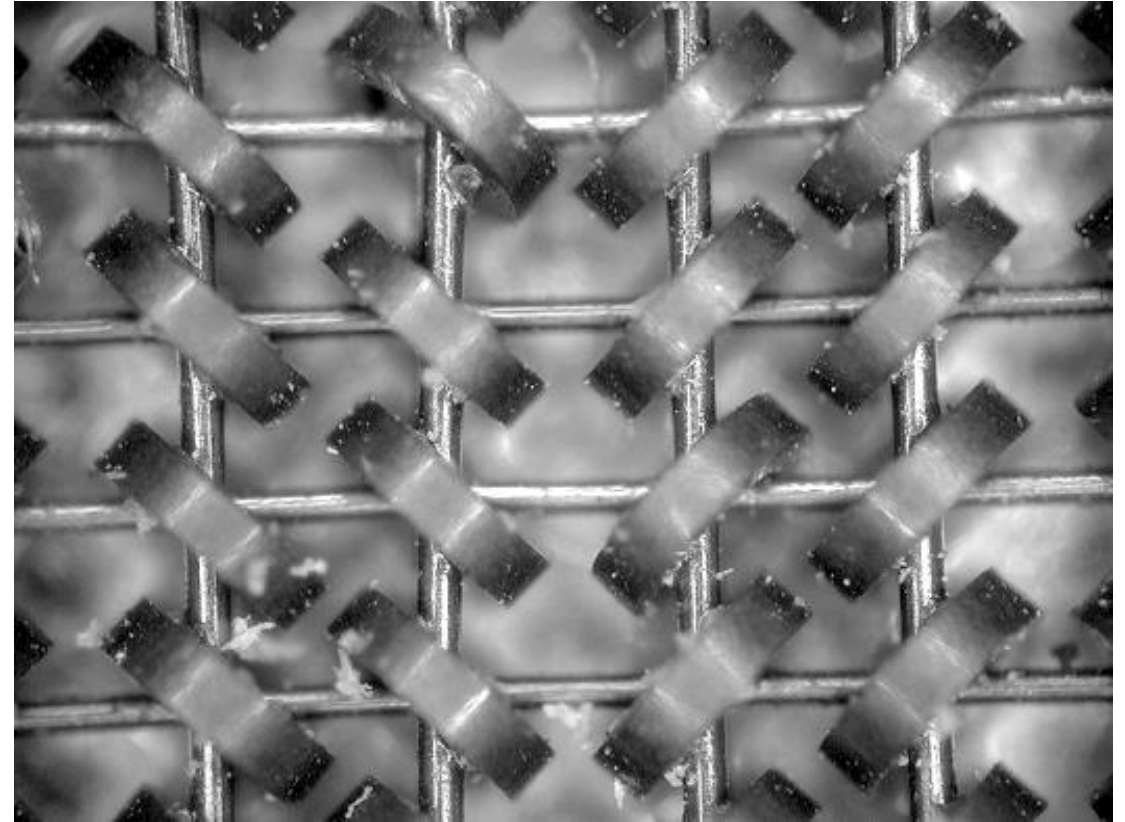


Memory – early systems

- The first computers used vacuum tubes to store data
- An alternative to vacuum tubes were relays
- Based on cathode ray tubes, the Williams tube was the first random access memory, capable of storing several thousands of bits, but only for some seconds

Memory – early systems

- The first truly useable type of main memory was magnetic core memory, introduced in 1951
- The dominant type of memory until the late 1960s
- Core memory was replaced by RAM chips in the 1970s



RAM memory

- RAM: Random Access Memory
 - Any piece of data stored in RAM can be read in the same amount of time, regardless of its physical location
- Based on transistor technology, typically implemented in large amounts in Integrated Circuits (ICs)
- Data is volatile – it remains available as long as the RAM is powered



RAM memory

- Static RAM (SRAM)
 - Uses flip-flop circuitry to store bits
 - Six transistors per bit
- Dynamic RAM (DRAM)
 - Uses a charge in a capacitor
 - One transistor per bit
 - DRAM loses its data after a short time due to the leakage of the capacitors
 - To keep data available in DRAM it must be refreshed regularly (typically 16 times per second)

DDR



System Bandwidth

2.1 – 3.2 GB/s

DDR2



4.2 – 6.4 GB/s

DDR3



8.5 – 14.9 GB/s

DDR4



17 – 21.3 GB/s

DDR5



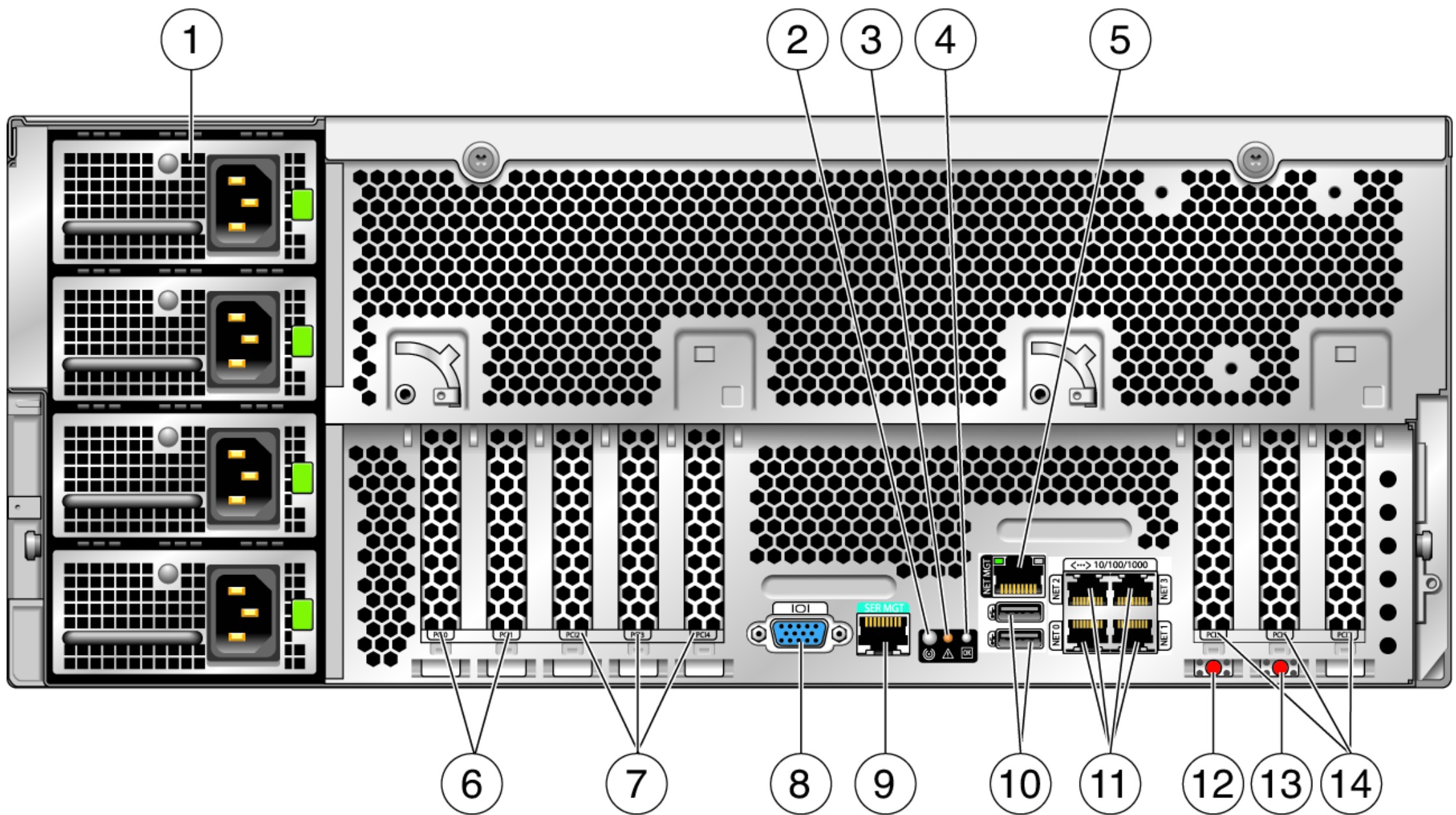
33.6 – 69.2 GB/s

BIOS

- The Basic Input/Output System (BIOS) is a set of instructions stored on a memory chip located on the computer's motherboard
- The BIOS controls a computer from the moment it is powered on, to the point where the operating system is started
- Mostly implemented in a Flash memory chip
- It is good practice to update the BIOS software regularly

Interfaces

- Connecting computers to external peripherals is done using interfaces
- External interfaces use connectors located at the outside of the computer case
- USB and Thunderbolt are mostly used today



14. PCIe card slots

Interfaces - USB

- The Universal Serial Bus (USB) was introduced in 1996 as a replacement for most of the external interfaces on servers and PCs
- Can provide operating power to attached devices
- Up to seven devices can be daisy-chained
 - Hubs can be used to connect multiple devices to one USB computer port
- In 2013, USB 3.1 was introduced
 - Provides a throughput of 10 Gbit/s
- In 2014, USB Type-C was introduced
 - Smaller connector
 - Ability to provide more power to connected devices, including powering laptops
 - In order to transport this much power, the voltage of the USB-C power charger is raised automatically to a maximum of 20 V
- The USB-C connector can also provide the USB 4 protocol, delivering up to 40 Gbit/s throughput

Interfaces - Thunderbolt

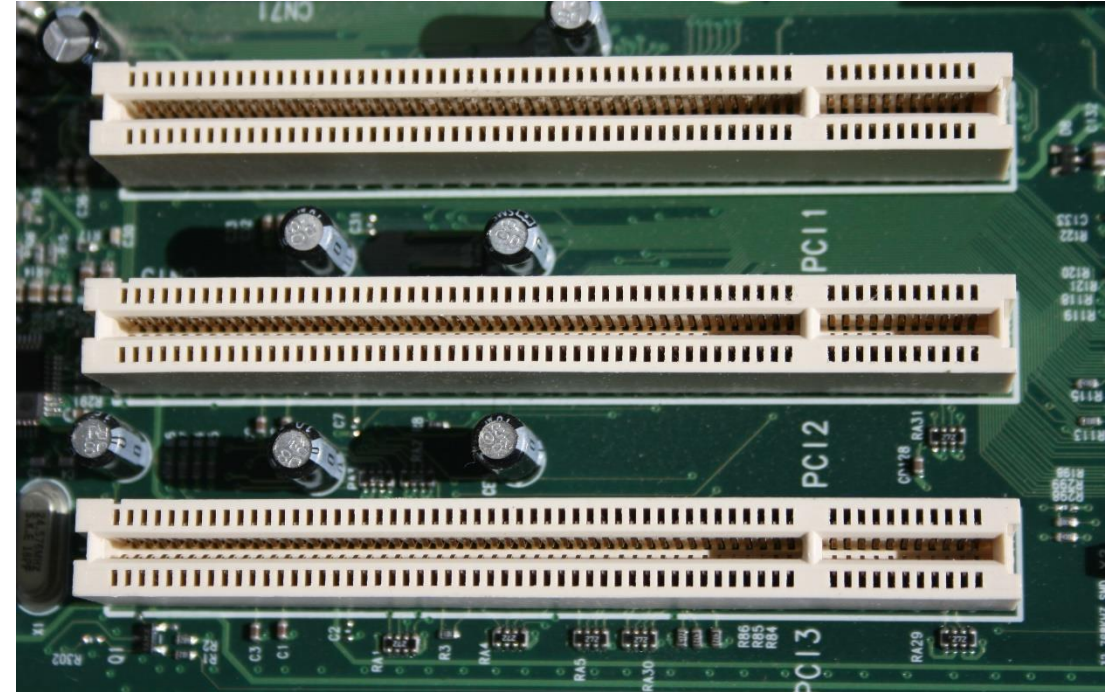
- Thunderbolt, also known as Light Peak, was introduced in 2011
- Thunderbolt 3 was released in 2015
 - Can provide a maximum throughput of 80 Gbit/s
 - Provide 100 W power to devices
 - Uses the USB Type-C connector
 - Backward compatible with USB 3.1



Source: <https://www.crest.com.au/blogs/learn/usb4-vs-usb3-whats-the-difference>

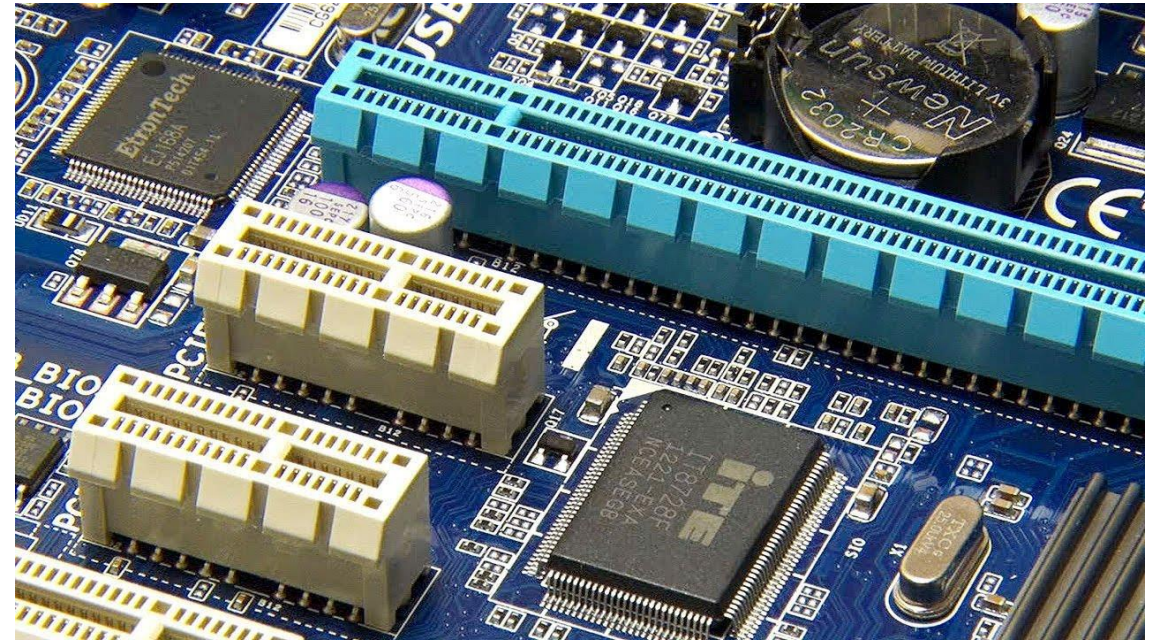
Interfaces - PCI

- Internal interfaces, typically some form of Peripheral Component Interconnect (PCI), are located on the system board of the computer, inside the case, and connect expansion boards like network adapters and disk controllers
- Uses a shared parallel bus architecture



Interfaces - PCIe

- PCI Express (PCIe) uses a topology based on point-to-point serial links, rather than a shared parallel bus architecture
- Routed by a hub on the system board acting as a crossbar switch
- Despite the availability of the much faster PCIe, conventional PCI remains a very common interface in computers



PCI and PCIe

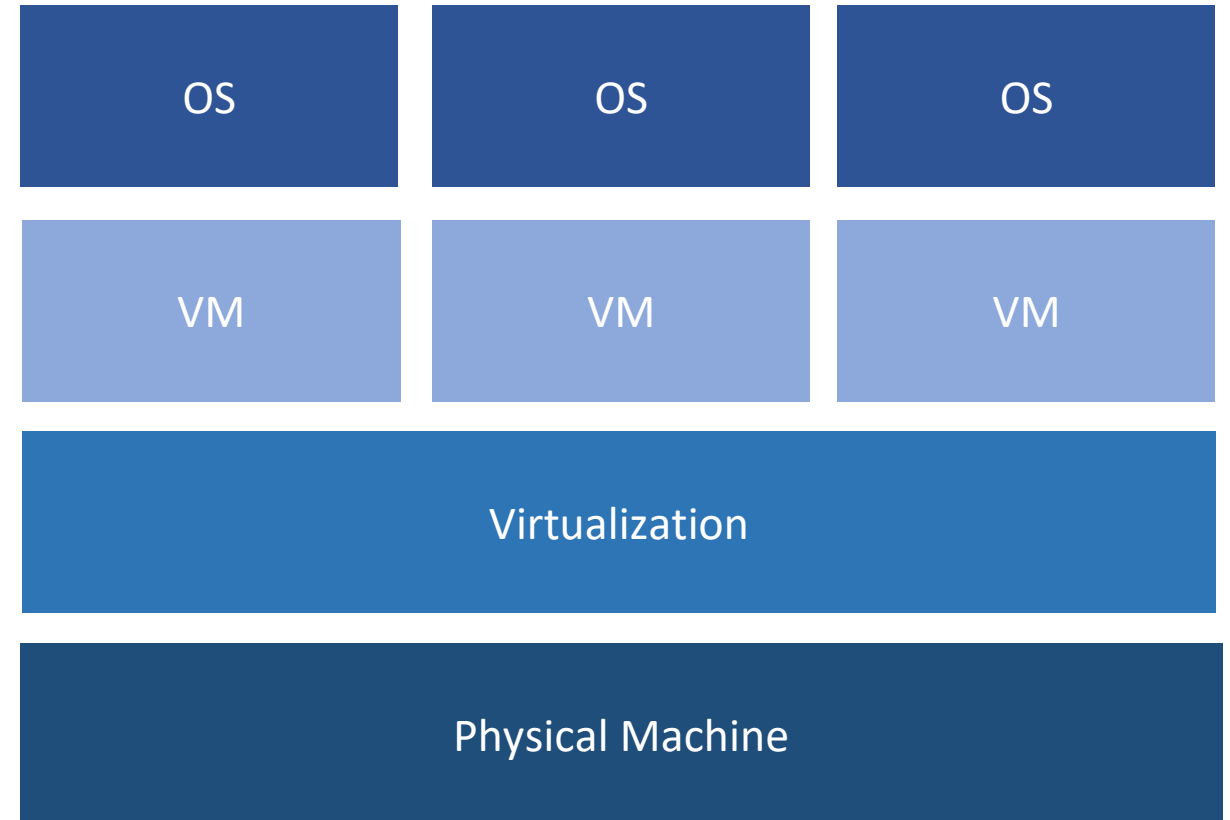
PCI speeds in Gbit/s

	Lanes				
	1	2	4	8	16
PCI 32-bit/33 MHz	0.1				
PCI 32-bit/66 MHz	0.3				
PCI 64-bit/33 MHz	0.3				
PCI 64-bit/66 MHz	0.5				
PCI 64-bit/100 MHz	0.8				
PCIe 1.0	0.2	0.5	1	2	4
PCIe 2.0	0.5	1	2	4	8
PCIe 3.0	1	2	4	8	16
PCIe 4.0	2	4	8	16	31
PCIe 5.0	4	8	16	32	63
PCIe 6.0	8	15	30	61	121
PCIe 7.0	15	30	61	121	242

Compute virtualization

Compute virtualization

- Compute virtualization is also known as Server virtualization/Software Defined Compute
- Introduces an abstraction layer between physical computer hardware and the operating system using that hardware



Compute virtualization

- A virtual machine is a logical representation of a physical computer in software
- New virtual machines can be provisioned without the need for a hardware purchase
- Costs can be saved on hardware, power, and cooling by consolidating many physical computers as virtual machines on fewer (bigger) physical machines

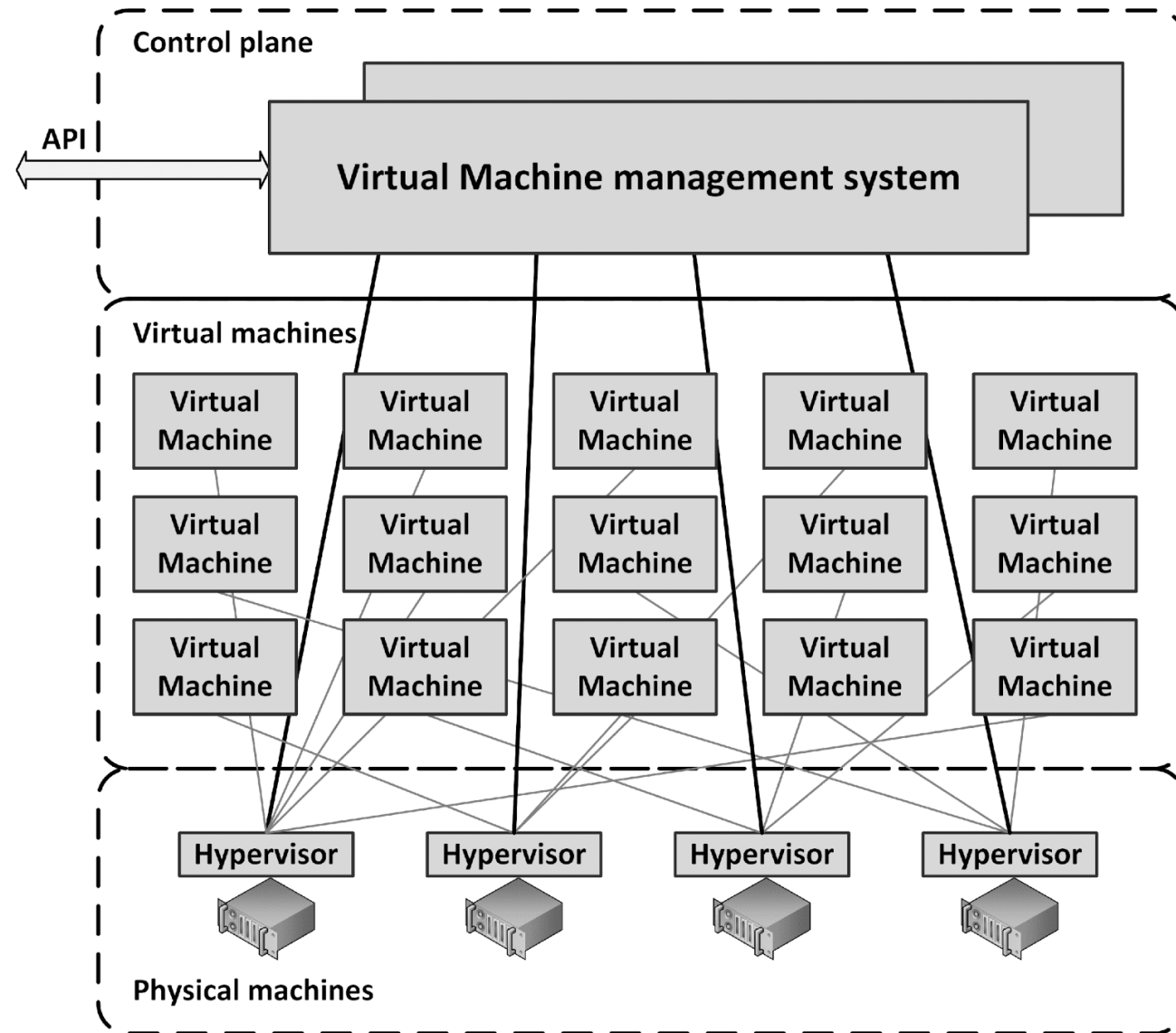
Compute virtualization and public cloud

- Virtualization allows users to create, start, and stop virtual machines (VMs) in a portal, CLI, or by calling an API
- Cloud-based VMs come in a variety of types, each with its own characteristics
- Naming can vary:
 - In Azure, VMs are simply called Virtual Machines
 - In GCP they are VM instances
 - In AWS they are called EC2 instances

Virtual Machine Management

- Virtual machines are typically managed using one redundant centralized virtual machine management system
 - Enables systems managers to manage more machines with the same number of staff
 - Allows managing the virtual machines using APIs
- All physical machines are running a hypervisor and all hypervisors are managed as one layer using management software

Virtual Machine Management



Virtual Machine Management

- Some virtualization platforms allow running virtual machines to be moved automatically between physical machines
- Benefits:
 - When a physical machine fails, all virtual machines that ran on the failed physical machine can be restarted automatically on other physical machines
 - Virtual machines can automatically be moved to the least busy physical machines
 - Some physical machines can get fully loaded while other physical machines can be automatically switched off, saving power and cooling cost
 - Enables hardware maintenance without downtime

Disadvantages of computer virtualization

- Because creating a new virtual machine is so easy, virtual machines tend to get created for all kinds of reasons
- This effect is known as "virtual machine sprawl"
- All VMs:
 - Must be managed
 - Use resources of the physical machine
 - Use power and cooling
 - Must be back-upped
 - Must be kept up to date by installing patches

Disadvantages of computer virtualization

- Introduction of an extra layer in the infrastructure
 - License fees
 - Systems managers training
 - Installation and maintenance of additional tools
- Virtualization cannot be used on all servers
 - Some servers require additional specialized hardware, like modem cards, USB tokens or some form of high speed I/O like in real-time SCADA systems
- Virtualization is not supported by all application vendors
 - When the application experiences some problem, systems managers must reinstall the application on a physical machine before they get support

Virtualization technologies

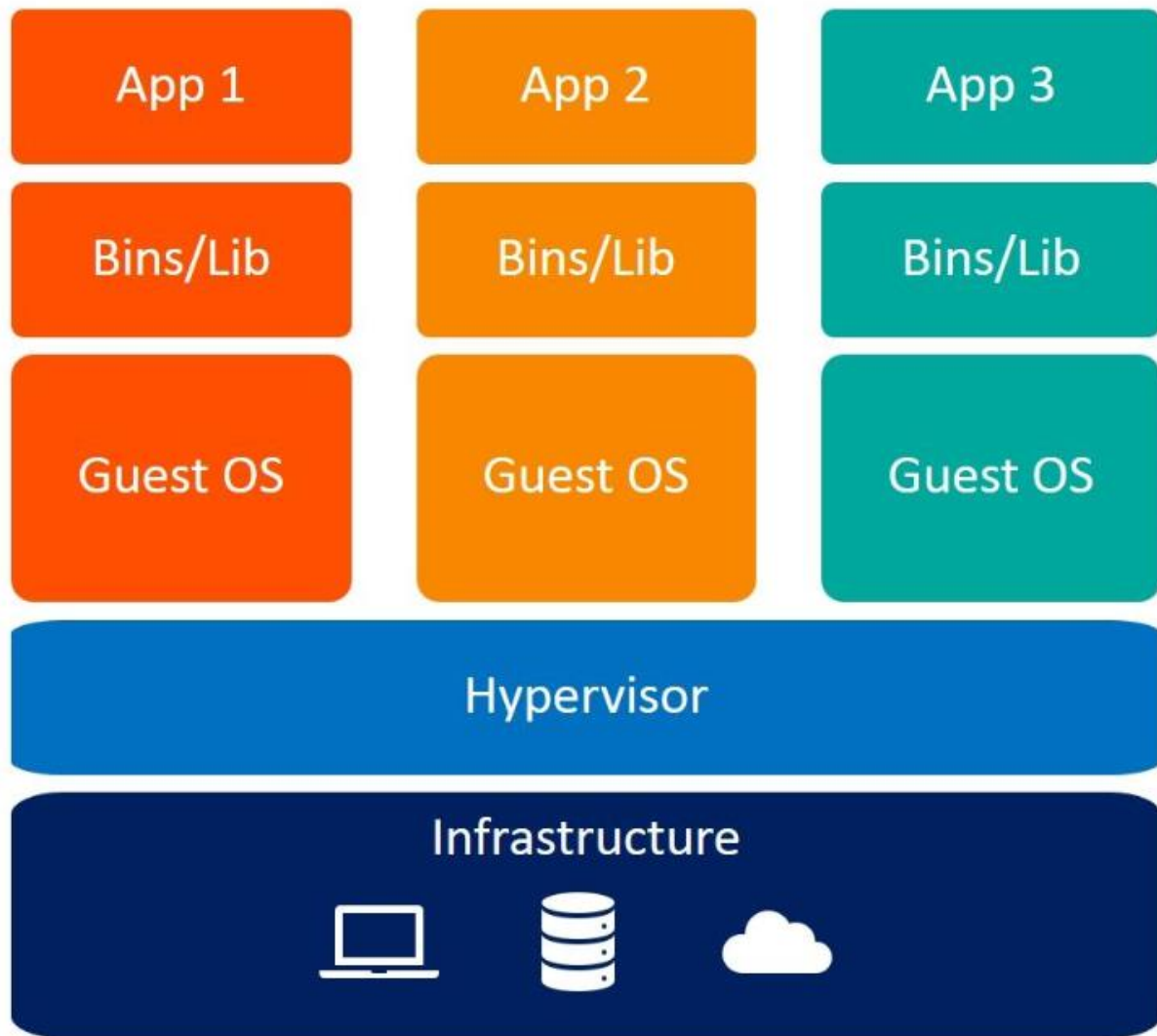
- Emulation
 - Can run programs on a computer, other than the one they were originally intended for
 - Run a mainframe operating system on a x86 server
- Logical Partitions (LPARs)
 - Hardware based
 - Used on mainframe and midrange systems

Virtualization technologies

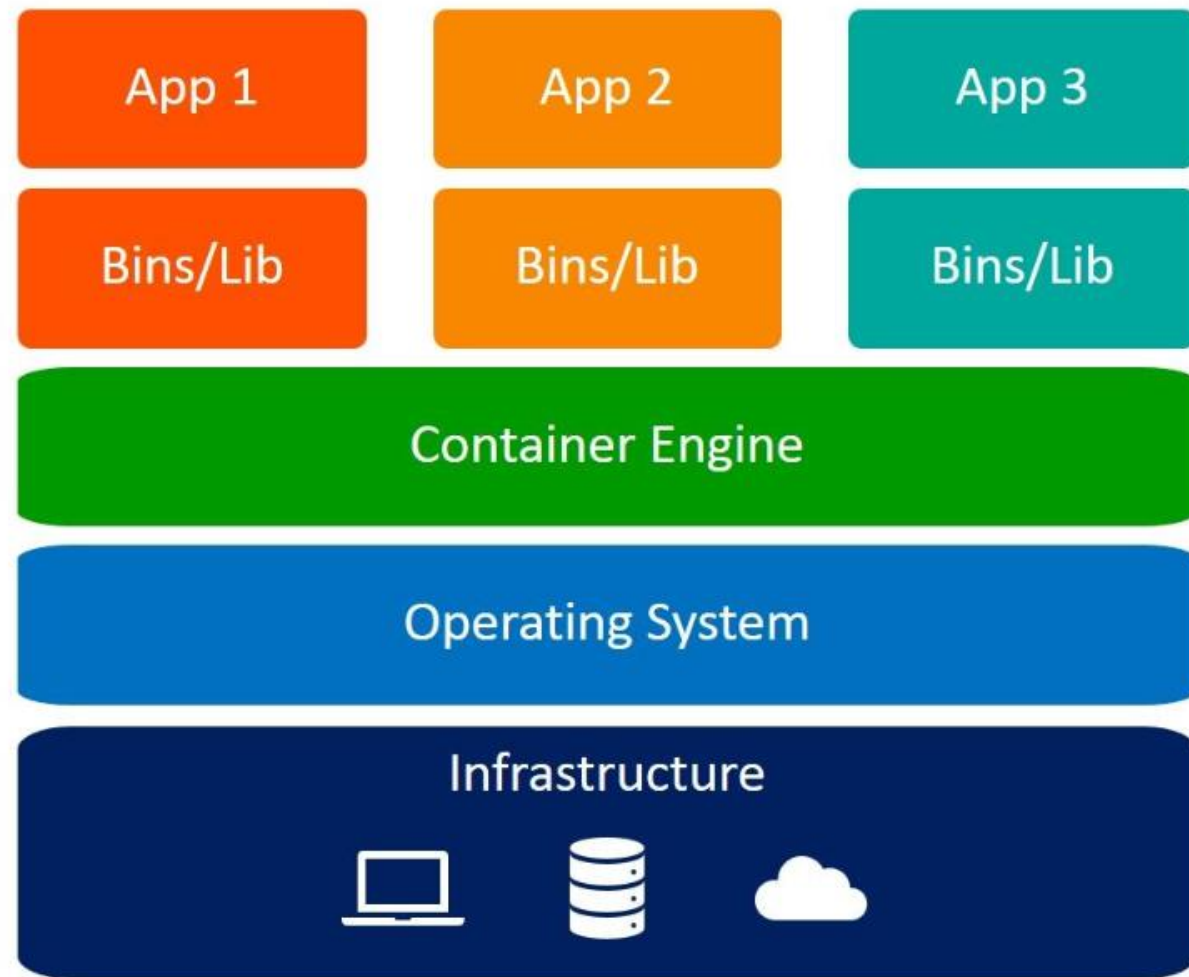
- Hypervisors
 - Control the physical computer's hardware and provide virtual machines with all the services of a physical system
 - Virtual CPUs
 - BIOS
 - Virtual devices
 - Virtualized memory management
 - Three types:
 - Binary translation
 - Paravirtualization
 - Hardware assisted virtualization (most used on x86 servers)

Container technology

- Container technology is a server virtualization method in which the kernel of an operating system provides multiple isolated user-space instances, instead of just one
- Containers look and feel like a real server from the point of view of its owners and users, but they share the same operating system kernel
- Containers are part of the Linux kernel since 2008
- Developers define the contents of containers



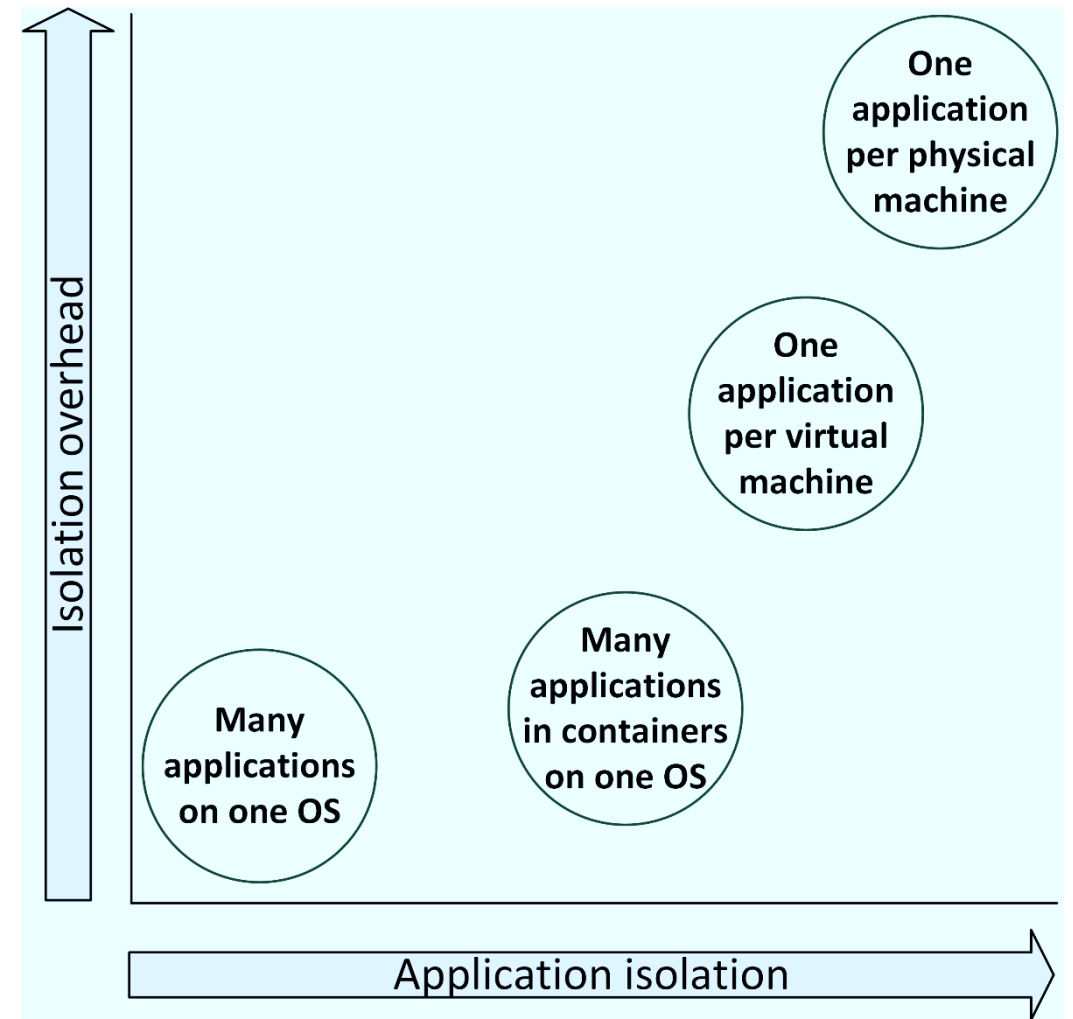
Virtual Machines



Containers

Container technology

- Containers are a balance between isolation and overhead of running isolated applications
- Virtual machines are isolated from the hardware using special CPU instructions
- Containers don't have that level of isolation



Container technology

- Containers have a number of benefits:
 - Isolation
 - Applications or application components can be encapsulated in containers, each operating independently and isolated from each other
 - Portability
 - Since containers typically contain all components the application needs to function, including libraries and patches, containers can be run on any infrastructure that is capable of running containers using the same kernel version
 - Easy deployment
 - Containers allow developers to quickly deploy new software versions, as their containers can be moved from the development environment to the production environment unaltered

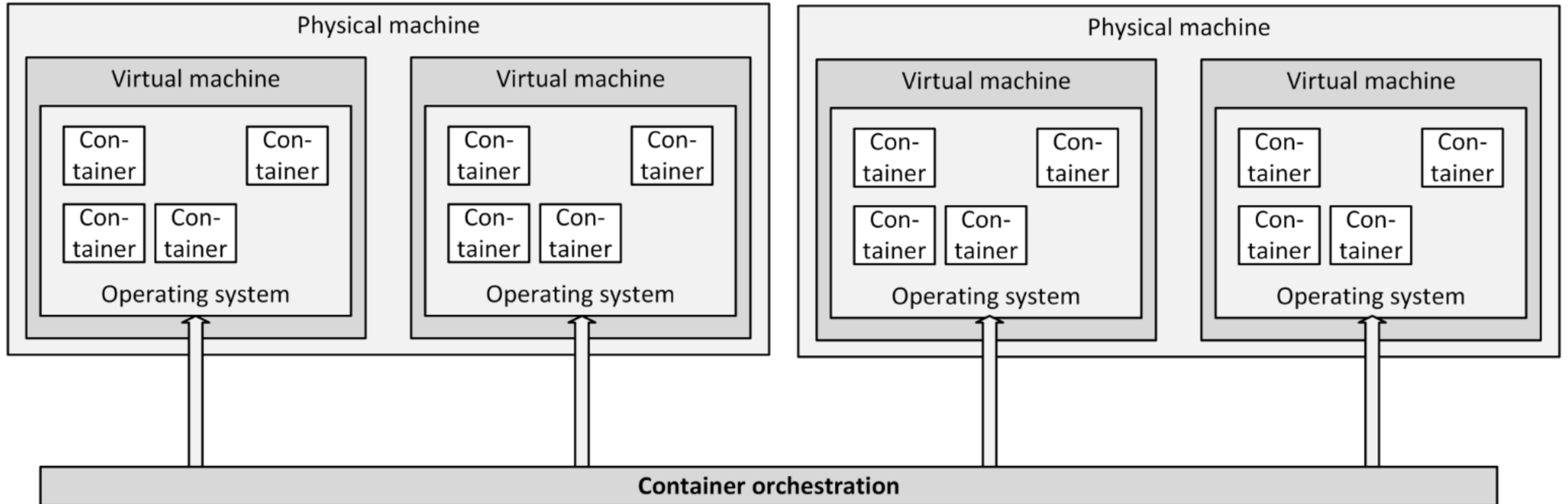
Container implementation

- Containers are based on 3 technologies that are all part of the Linux kernel:
 - Chroot (also known as a jail)
 - Changes the root directory for the current running process
 - Ensures processes cannot access files outside the designated directory tree
 - Namespaces
 - Allows complete isolation of an applications' view of the operating environment
 - Process trees, networking, user IDs and mounted file systems
 - Cgroups
 - Limits and isolates the resource usage of a collection of processes
 - PU, memory, disk I/O, network
- Linux Containers (LXC), introduced in 2008, is a combination of these
- Docker is a popular implementation of a container ecosystem

Container orchestration

- Container orchestration abstracts the resources of a cluster of machines and provides services to containers
- A container orchestrator enables containers to be run anywhere on a cluster of machines
 - Schedules the containers to run on any machine that has resources available
 - Acts like a kernel for the combined resources of an entire datacenter

Container orchestration



Container orchestration

- Kubernetes (also spelled as K8S) is an open source project hosted by the Cloud Native Computing Foundation (CNCF), based on Google's internal Borg System
- Kubernetes provides:
 - **Replication**: to deploy multiple instances of an application
 - **Load balancing and service discovery**: to route traffic to these replicated containers
 - **Basic health checking and repair**: to ensure a self-healing system
 - **Scheduling**: to group many machines into a single pool and distribute work to them
- Public cloud Kubernetes implementations:
 - Amazon Elastic Kubernetes Service (EKS)
 - Azure Kubernetes Services (AKS)
 - Google Kubernetes Engine (GKE)

Serverless computing

- In serverless computing, application source code can be run directly by a cloud provider
- Serverless computing allows developers to build and run applications and services without having to manage infrastructure
- The provider charges based on the actual number of resources used by an application, rather than charging for pre-allocated amounts of computing resources

Serverless computing

- All major public cloud providers offer serverless computing:
 - AWS has Lambda
 - Azure has Azure Functions
 - GCP has Google Cloud serverless
- Serverless locks the organization into the platform, as serverless computing often uses special features that are only available on the specific cloud platform

Mainframes

- A mainframe is a high-performance computer made for high-volume, I/O-intensive computing
 - Expensive
 - Mostly used for administrative processes
 - Optimized for handling high volumes of data
- IBM is the largest vendor – it has 90% market share
- The end of the mainframe is predicted for decades now, but mainframes are still widely used
- Today's mainframes are still large (the size of a few 19" racks), but they don't fill-up a room anymore

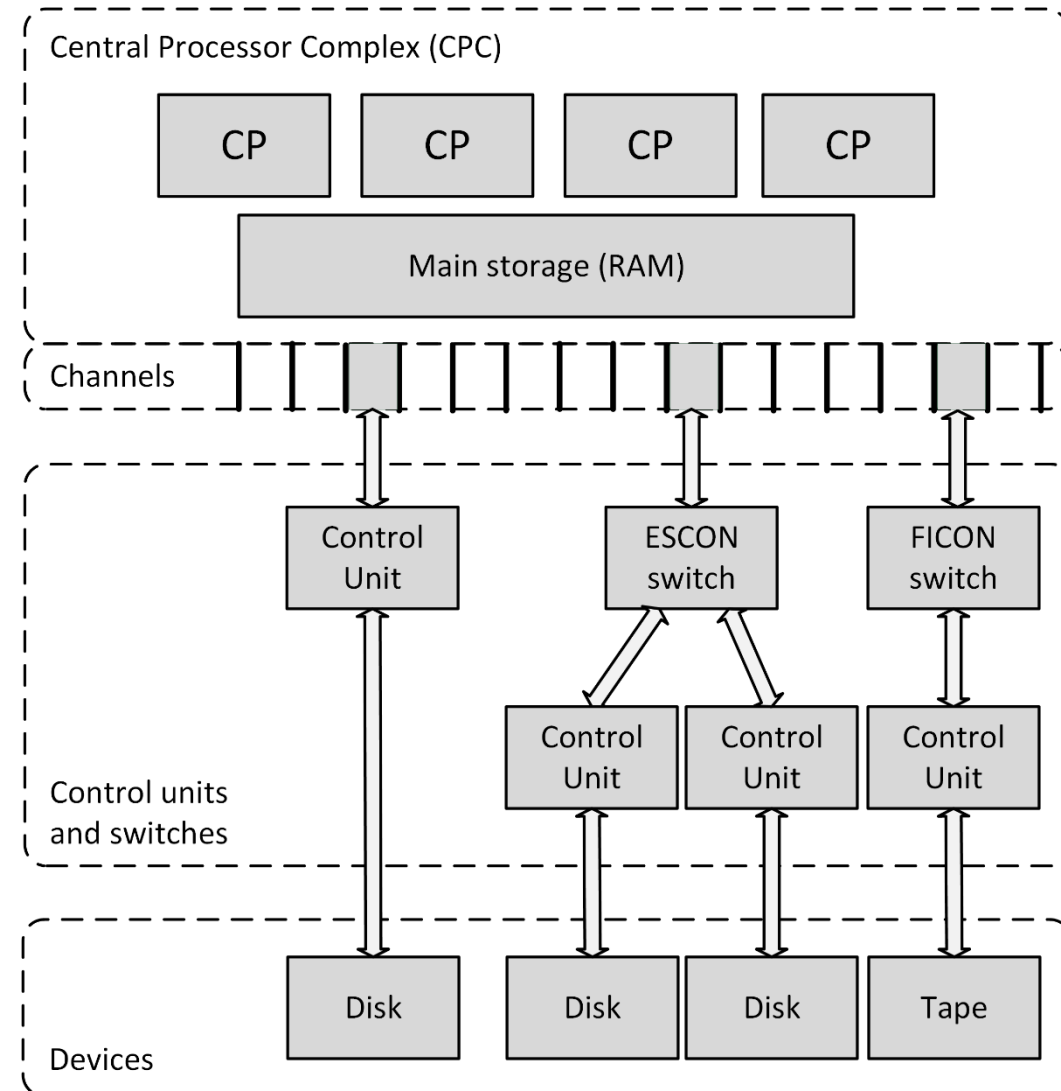
Mainframes

- Mainframes are highly reliable, typically running for years without downtime
- Much redundancy is built in
 - Hardware can be upgraded and repaired while the mainframe is operating without downtime
- The latest IBM z16 mainframe:
 - Up to 10TB of memory
 - Up to 200 processors
 - Running at a 5GHz clock speed
 - Can run up to 8000 virtual machines simultaneously



Mainframe architecture

- A mainframe consists of:
 - Processing units (PUs)
 - Memory
 - I/O channels
 - Control units
 - Devices, all placed in racks (frames)



Mainframe architecture – PU, memory, and disks

- In the mainframe world, the term PU (Processing Unit) is used instead of CPU
 - A mainframe has multiple PUs, so there is no central processing unit
 - The total of all PUs in a mainframe is called a Central Processor Complex (CPC)
- Each book package in the CPC cage contains from four to eight memory cards
 - For example, a fully loaded z16 mainframe has four book packages that can provide up to a total of 10 TB of memory
- Disks in mainframes are called DASD (Direct Attached Storage Device)
 - Comparable to a SAN in a midrange or x86 environment

Mainframe architecture – Channels and control units

- A channel provides a data and control path between I/O devices and memory
- Today's largest mainframes have 1024 channels
- A control unit is similar to an expansion card in an x86 or midrange system
 - Contains logic to work with a particular type of I/O device, like a printer or a tape drive

Mainframe architecture – Channels and control units

- Channel types:
 - OSA
 - Connectivity to various industry standard networking technologies, including Ethernet
 - FICON
 - The most flexible channel technology, based on fiber-optic technology
 - With FICON, input/output devices can be located many kilometers from the mainframe to which they are attached
 - ESCON
 - An earlier type of fiber-optic technology
 - Almost as fast as FICON channels, but at a shorter distance

Mainframe virtualization

- Mainframes were designed for virtualization from the start
- Logical partitions (LPARs) are the default virtualization solution
- LPARs are equivalent to separate mainframes
- A common number of LPARs in use on a mainframe is less than ten
- The mainframe operating system running on each LPAR is designed to concurrently run a large number of applications and services, and can be connected to thousands of users at the same time
- Often one LPAR runs all production tasks while another runs the consolidated test environment

Midrange systems

- The midrange platform is positioned between the mainframe platform and the x86 platform
- Built using parts from only one vendor, and run an operating system provided by that same vendor
- This makes the platform:
 - Stable
 - High available
 - Secure

Midrange systems

- Today midrange systems are produced by three vendors:
 - IBM
 - Power Systems series
 - Operating system: AIX UNIX, Linux and IBM i
 - Hewlett-Packard
 - HP Integrity systems
 - Operating system: HP-UX UNIX and OpenVMS
 - Oracle
 - Sun Microsystems's based SPARC servers
 - Operating system: Solaris UNIX

Midrange systems - History

- Minicomputer evolved in the 1960s as small computers that became possible with the use of IC and core memory technologies
- Small was relative
 - A single minicomputer typically was housed in a few cabinets the size of a 19" rack
- The first commercially successful minicomputer was DEC PDP-8, launched in 1964



Midrange systems - History

- Minicomputers became powerful systems
 - They ran full multi-user, multitasking operating systems like OpenVMS and UNIX
- In the 1980s, minicomputers (a.k.a. midrange systems) became less popular
 - A result of the lower cost of PCs, and the emergence of LANs
- Still used in places where high availability, performance, and security are very important

Midrange systems - Architecture

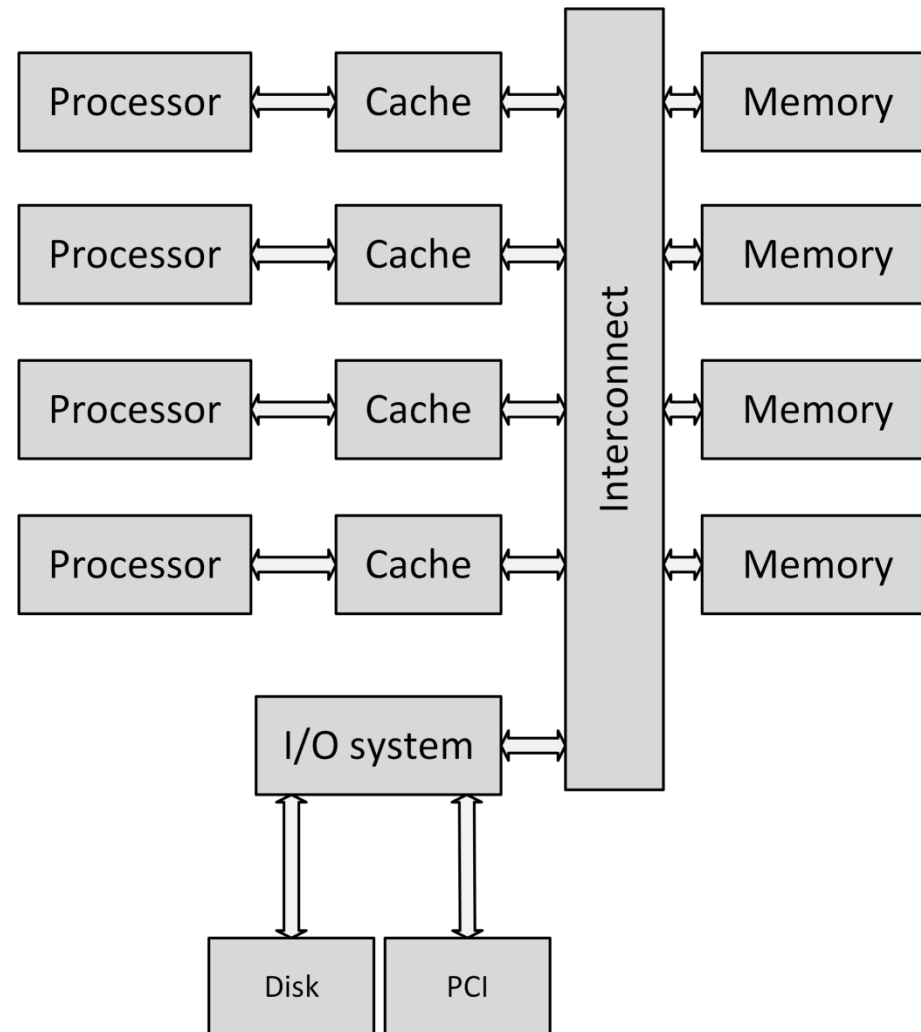
- The architecture of most midrange systems:
 - Uses multiple CPUs
 - Based on a shared memory architecture
- In a shared memory architecture, all CPUs in the system can access all installed memory blocks
 - Changes made in memory by one CPU are immediately seen by all other CPUs
 - A shared bus connects all CPUs and all RAM
 - The I/O system is also connected to the interconnection network

Midrange systems - Architecture

- Shared memory architectures come in two flavors:
 - Uniform Memory Access (UMA)
 - Non-Uniform Memory Access (NUMA)
 - Cache coherence is needed
 - If one CPU writes to a location in shared memory, all other CPUs must update their caches to reflect the changed data
 - Cache coherent versions are known as ccUMA and ccNUMA

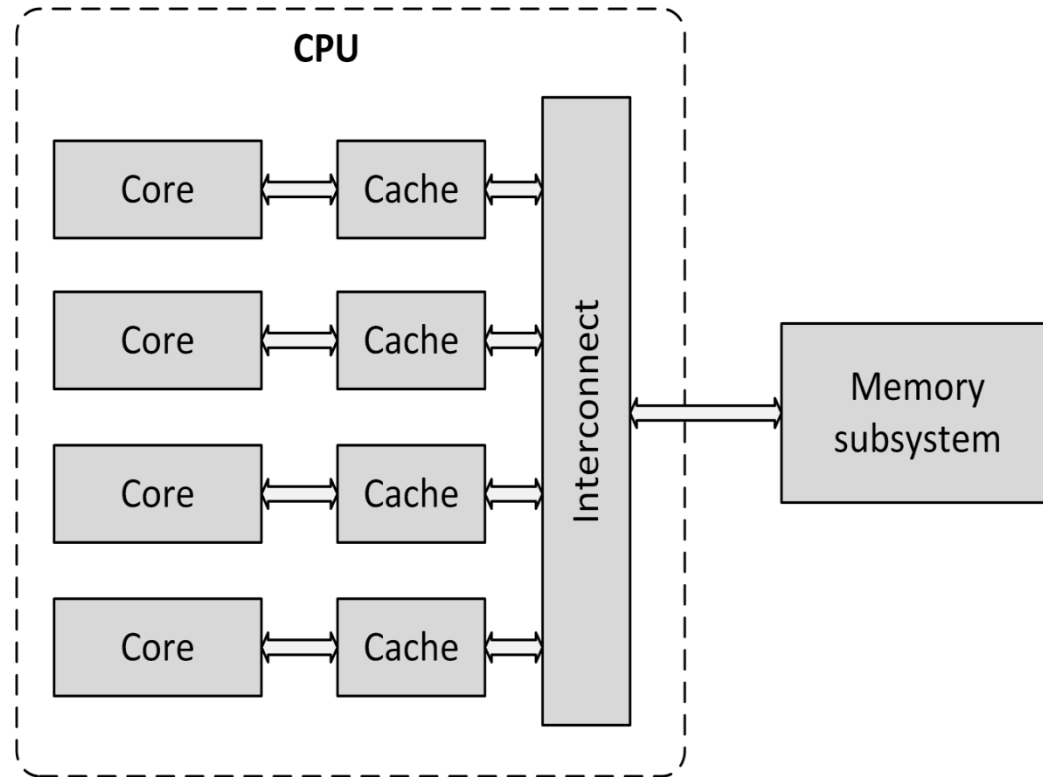
Midrange systems – UMA architecture

- The UMA architecture is one of the earliest styles of multi-CPU architectures, typically used in systems with no more than 8 CPUs
- The machine is organized into a series of nodes containing either a processor, or a memory block
- Nodes are interconnected, usually by a shared bus



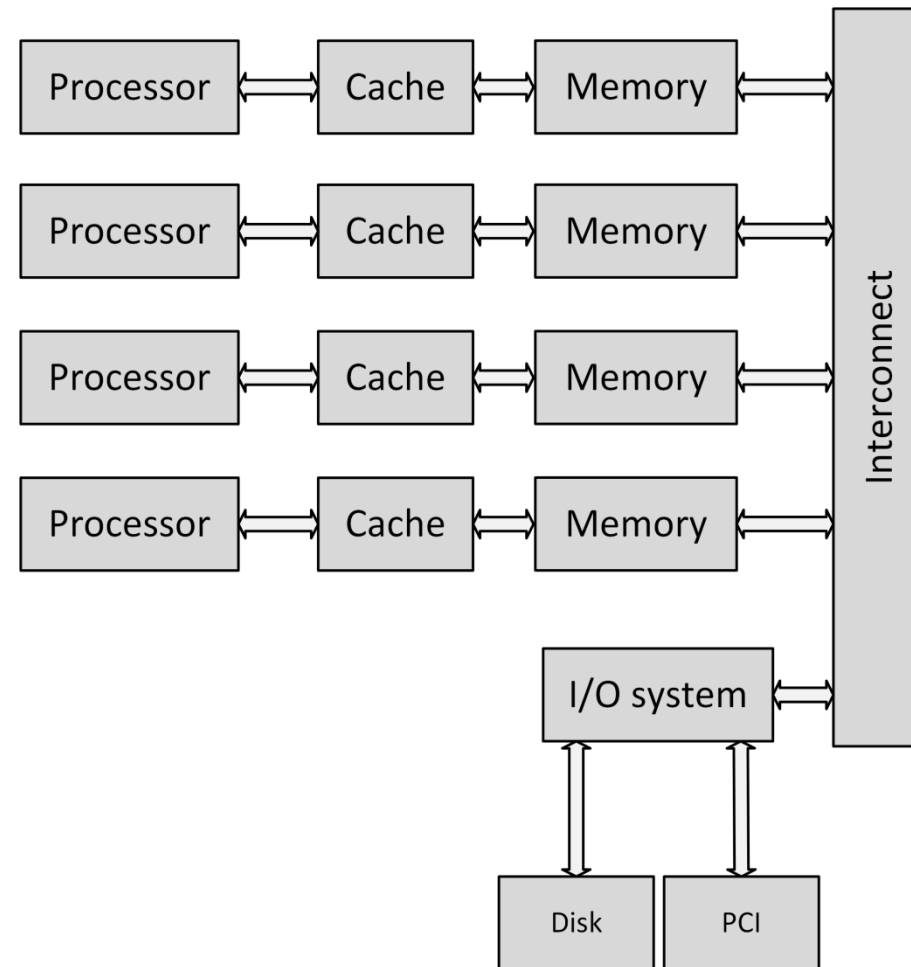
Midrange systems – SMP architecture

- UMA systems are also known as Symmetric Multi-Processor (SMP) systems
- SMP is used in x86 servers as well as early midrange systems
- Can be Implemented inside multi-core CPUs
 - The interconnect is implemented on-chip in the CPU
 - A single path to the main memory is provided between the chip and the memory subsystem



Midrange systems – NUMA architecture

- NUMA is a computer architecture in which the machine is organized into a series of nodes
- Each node contains processors and memory
- Nodes are interconnected using a crossbar interconnect
- When a processor accesses memory not within its own node, data must be transferred over the interconnect
 - Slower than accessing local memory
 - Memory access times are non-uniform



Midrange virtualization

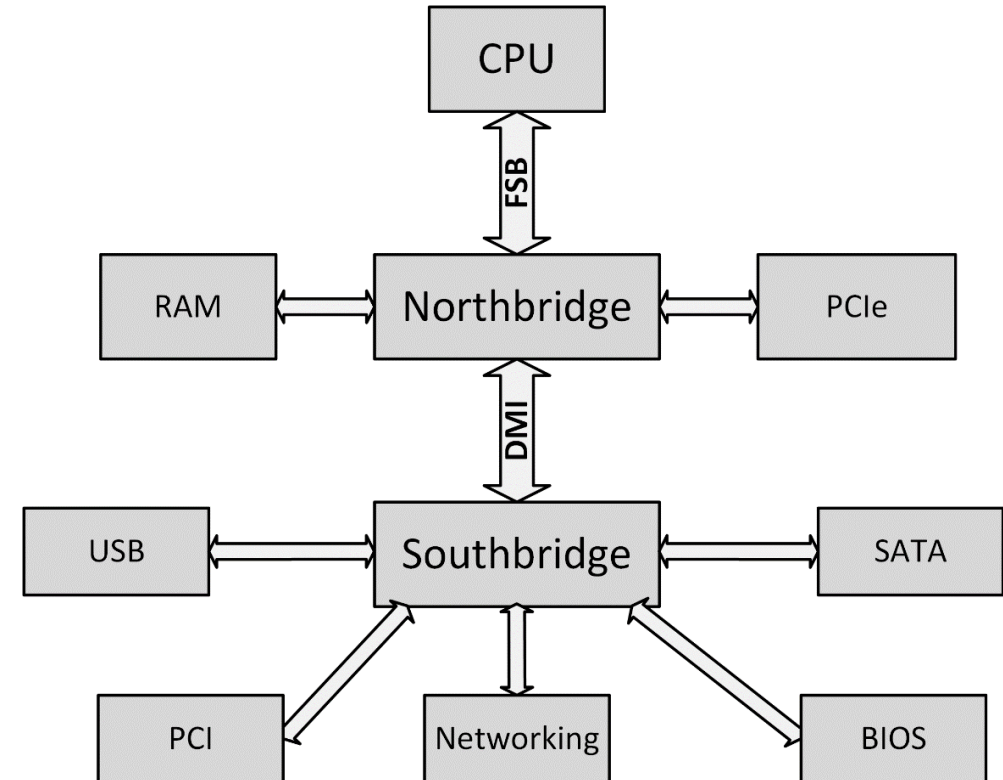
- Most midrange platform vendors provide virtualization based on LPARs
- LPARS are a type of hardware partitioning
 - IBM AIX: Workload/Working Partitions (WPARs)
 - HP: nPARs
 - Oracle Solaris: zones and containers

x86 servers

- The x86 platform is the most dominant server architecture today
- In the 1990s, x86 servers first started to appear.
 - They were basically big PCs, housed in 19" racks without dedicated keyboards and monitors
- x86 servers are produced by many vendors, like:
 - HP
 - Dell
 - Lenovo
 - HDS (Hitachi Data Systems)
 - Huawei
- Implementation of the platform is highly dependent on the vendor and the components available at a certain moment
- Most used operating systems are Microsoft Windows and Linux

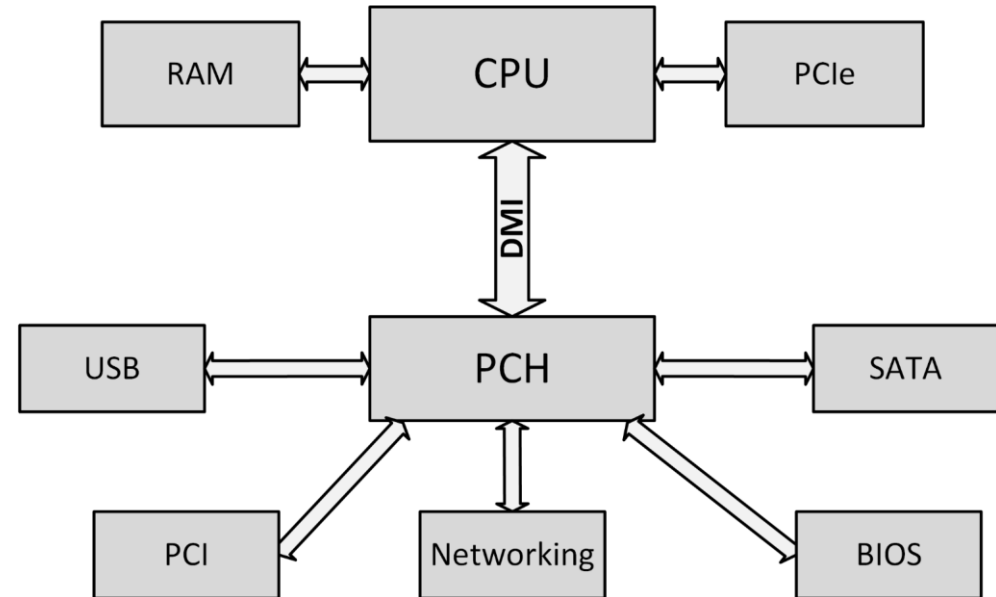
x86 servers - Architecture

- x86 architectures are defined by a CPU from the x86 family, and building blocks, integrated in a number of specialized chips, known as an x86 chipset
- Earlier x86 systems utilized a Northbridge / Southbridge architecture
 - Front Side Bus (FSB)
 - Direct Media Interface (DMI)



x86 servers - Architecture

- In the PCH architecture, the RAM and PCIe data paths are directly connected to the CPU
- The Northbridge is integrated in the CPU
- Since 2017 CPUs are a full system on a chip (SoC)
- SOC directly expose:
 - PCIe lanes
 - SATA
 - USB
 - High Definition Video



x86 virtualization

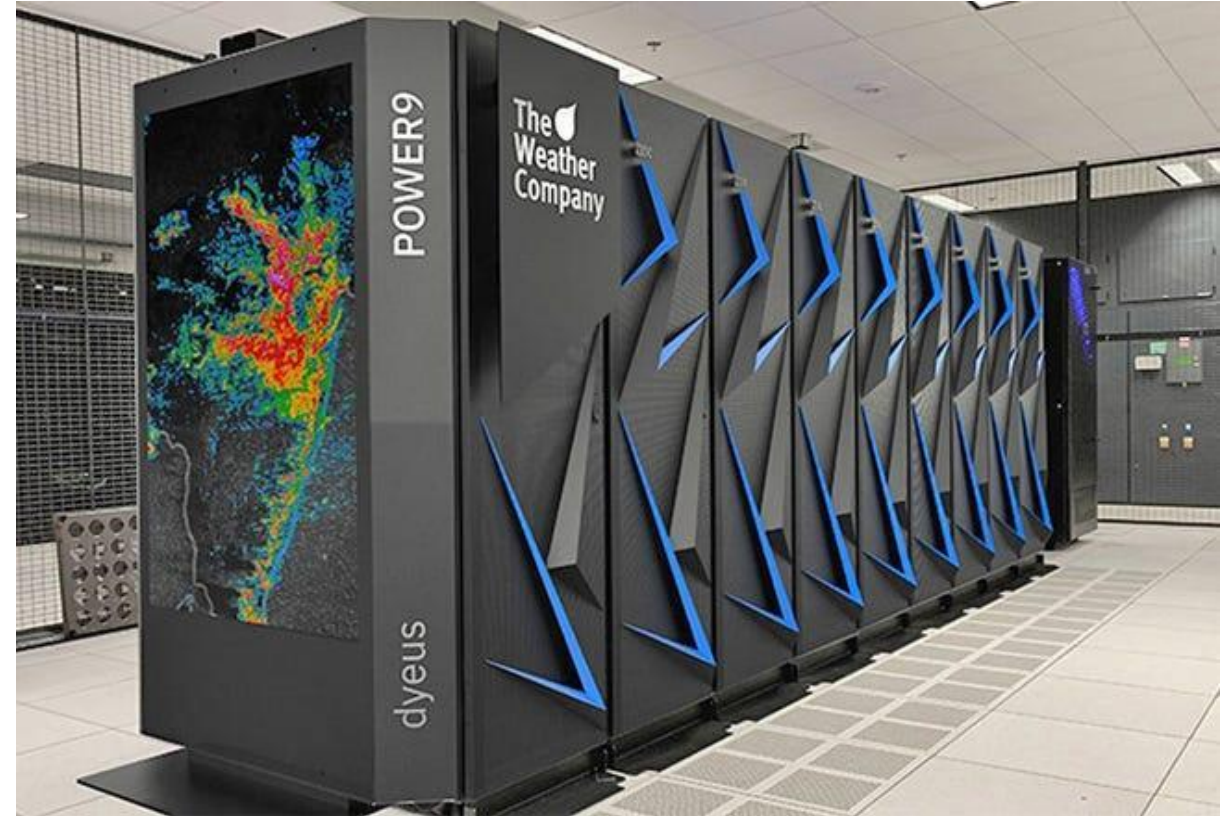
- On x86 platforms, most servers only run one application each
 - A Windows server running Exchange will probably not also run SharePoint
 - A Linux server running Apache will probably not also run MySQL
 - This is the main reason x86 systems use their hardware much less efficiently than midrange systems
- By running multiple operating systems – each in one virtual machine – on a large x86 server, resource utilization can be improved
- The most used products for virtualization on the x86 platform are:
 - VMware vSphere
 - Microsoft's Hyper-V
 - Citrix XenServer
 - Oracle VirtualBox
 - Red Hat RHEV

Supercomputers

- A supercomputer is a computer architecture designed to maximize calculation speed
 - This in contrast with a mainframe, which is optimized for high I/O throughput
- Supercomputers are the fastest machines available at any given time
- Used for highly compute-intensive tasks requiring floating point calculations, like:
 - Weather forecast calculations
 - Oil reservoir simulations
 - Rendering of animation movies

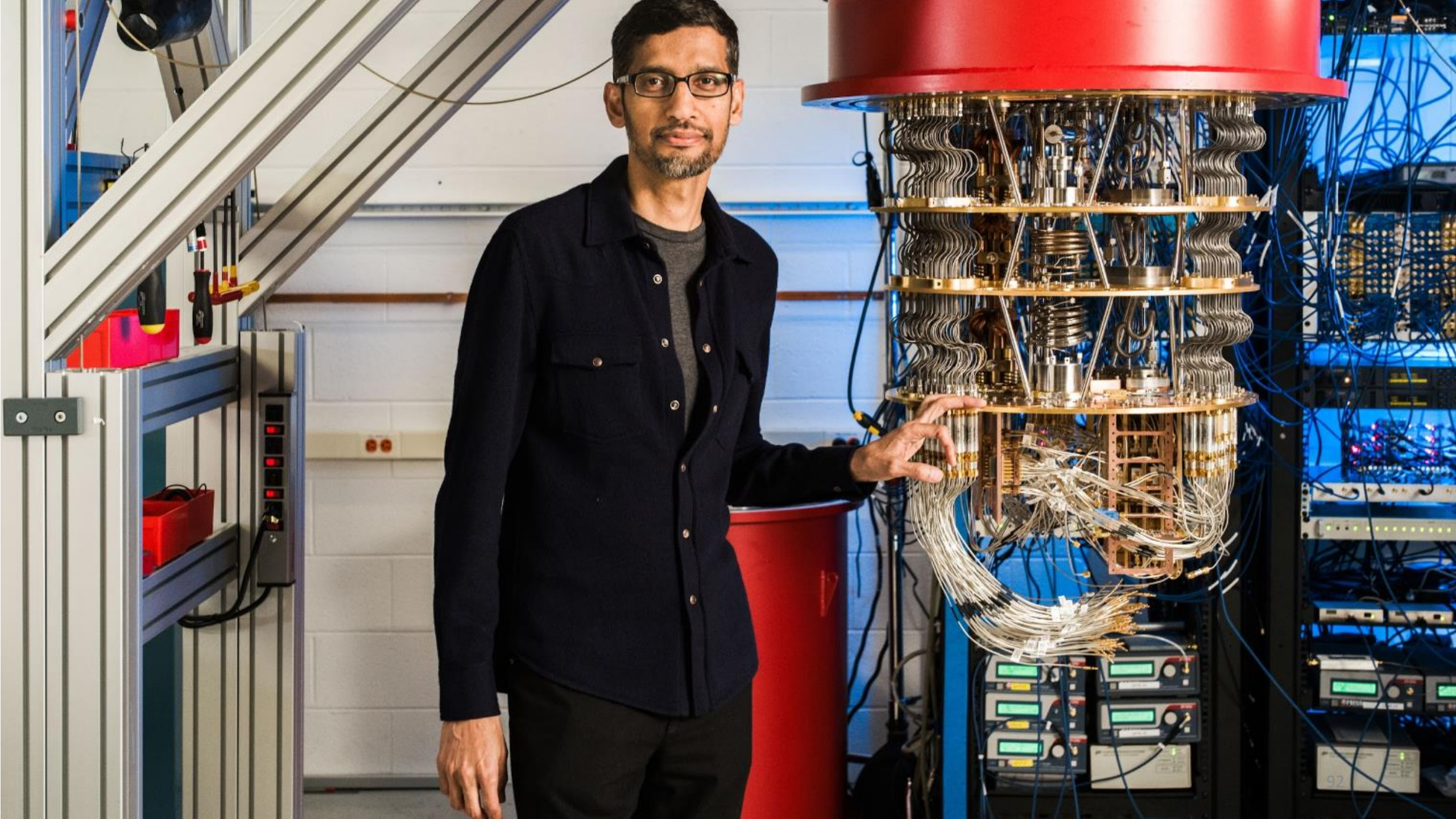
Supercomputers

- Originally, supercomputers were produced primarily by a company named Cray Research
 - Cray-1 (1976): 250 MFLOPS (Million Floating Point Operations per second)
 - Cray-2 (1985): 1,900 MFLOPS
- Nowadays high performance computing is done mainly with large arrays of x86 systems
 - The current Ryzen 7950x Zen4 CPU has a peak performance of 1011 GFLOPS; more than 530 times the performance of the Cray-2
- The fastest computer array is a cluster with 8,730,112 CPU cores, calculating at 1,685,650,000 GFLOPS, running Linux (top500.org)



Quantum computers

- A Quantum computer is a computer based on quantum mechanics
- They work by using tiny particles called qubits, which can be in multiple states at once.
- It's a strange and counterintuitive world where things can exist in multiple states at once and can be connected across vast distances.



Compute availability

Hot swappable components

- Hot swappable components are server components that can be installed, replaced, or upgraded while the server is running
- The virtualization and operating systems using the server hardware must be aware that components can be swapped on the fly



Parity memory

- To detect memory failures, parity bits can be used as the simplest form of error detecting code
- Parity bits enable the detection of data errors
- They cannot correct the error, as it is unknown which bit has flipped

DATA	PARITY
------	--------

1001 0110	0
-----------	---

1011 0110	1
-----------	---

0001 0110	0 -> ERROR: parity bit should have been 1!
-----------	--

ECC memory

- ECC memory not only detects errors, but is also able to correct them
- ECC Memory chips use Hamming Code or Triple Modular Redundancy (TMR) as the method of error detection and correction
- Memory errors are proportional to the amount of RAM in a computer as well as the duration of operation. Since servers typically contain many GBs of RAM and are in operation 24 hours a day, the likelihood of memory errors is relatively high and hence they require ECC memory

Virtualization availability

- All virtualization products provide failover clustering
 - When a physical machine fails, the virtual machines running on that physical machine can be configured to restart automatically on other physical machines
 - When a virtual machine crashes, it can be restarted automatically on the same physical machine

Virtualization availability

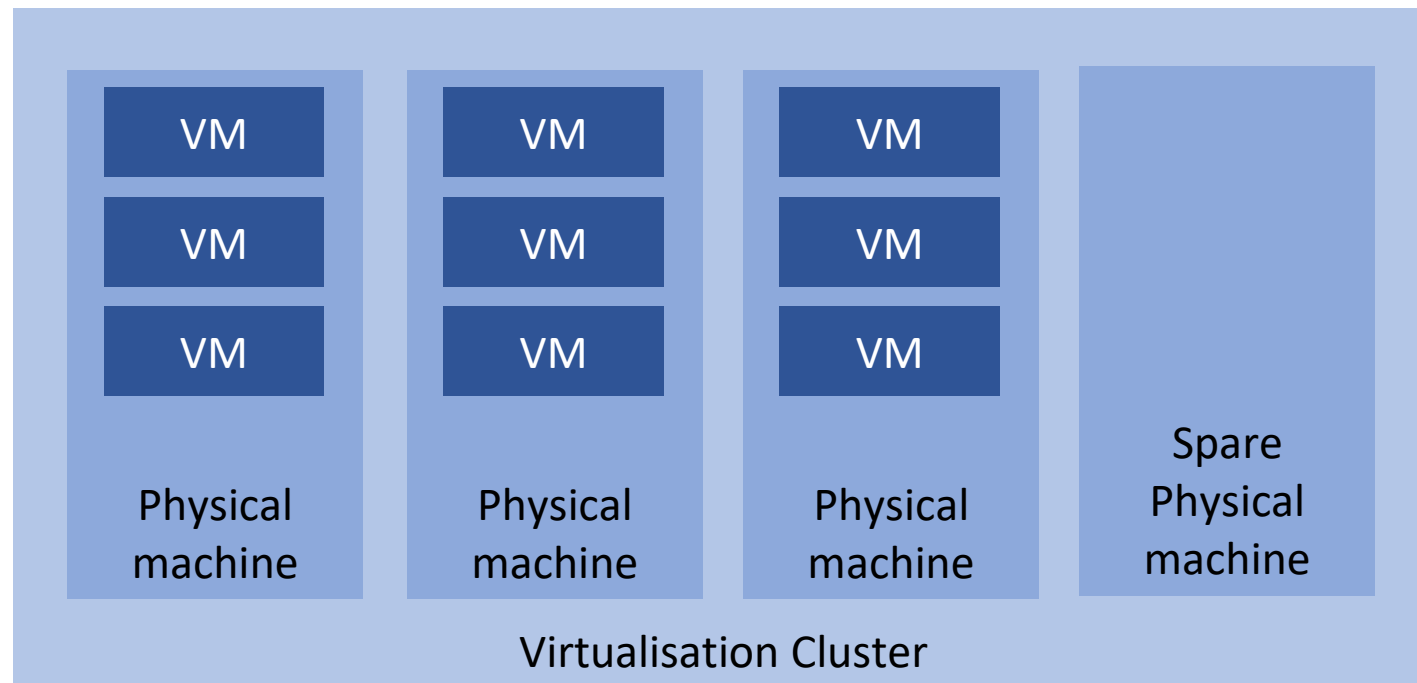
- The virtualization layer has no knowledge of the applications running on the virtual machine's operating system
- Failover clustering on virtualization level can only protect against two situations:
 - A **physical hardware failure**.
 - An **operating system crash** in a virtual machine

Virtualization availability

- To cope with the effects of a failure of a physical machine, a spare physical machine is needed
 - All hypervisors are placed in a virtualization cluster
 - The hypervisors on the physical machines check the availability of the other hypervisors in the cluster
 - One physical machine is running as a spare to take over the load of any failing physical machine

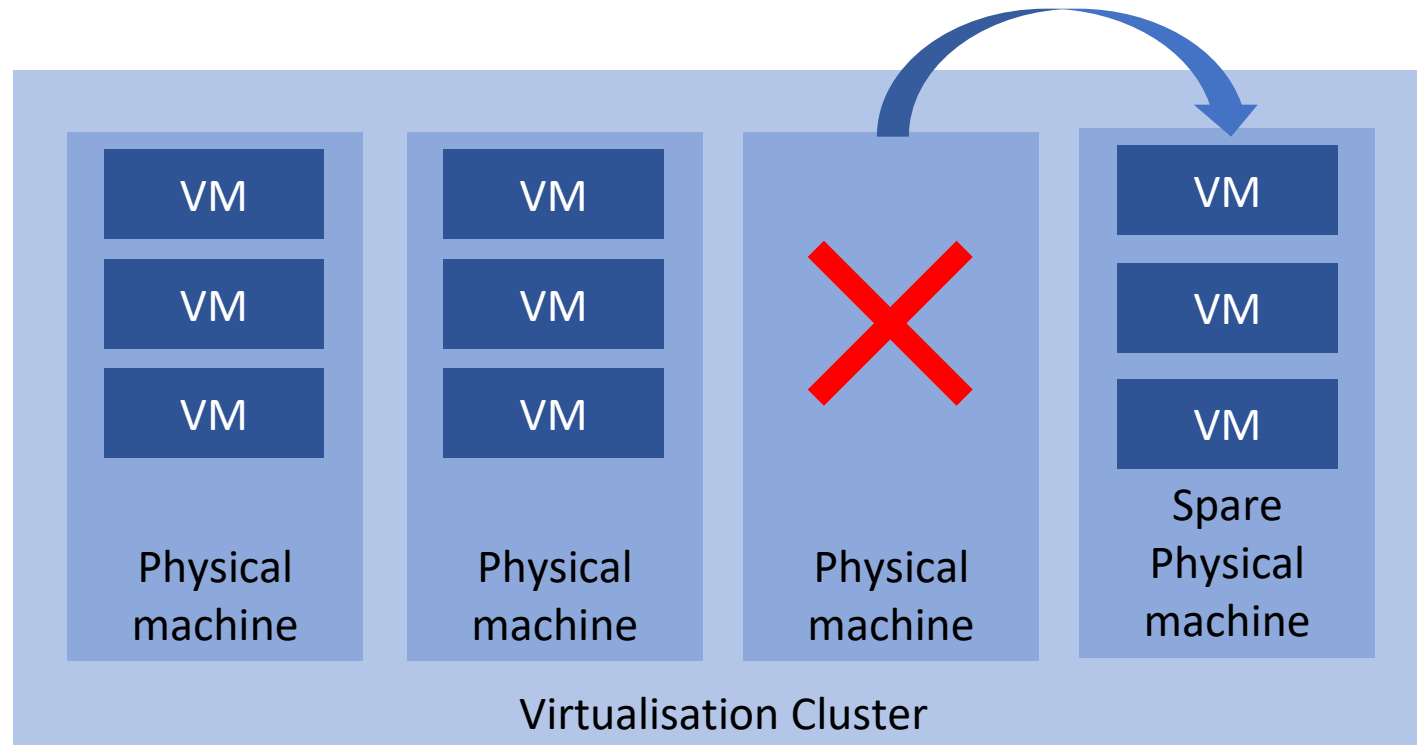
Virtualization availability

- When a physical machine fails, the virtual machines running on it are automatically restarted on the spare physical machine



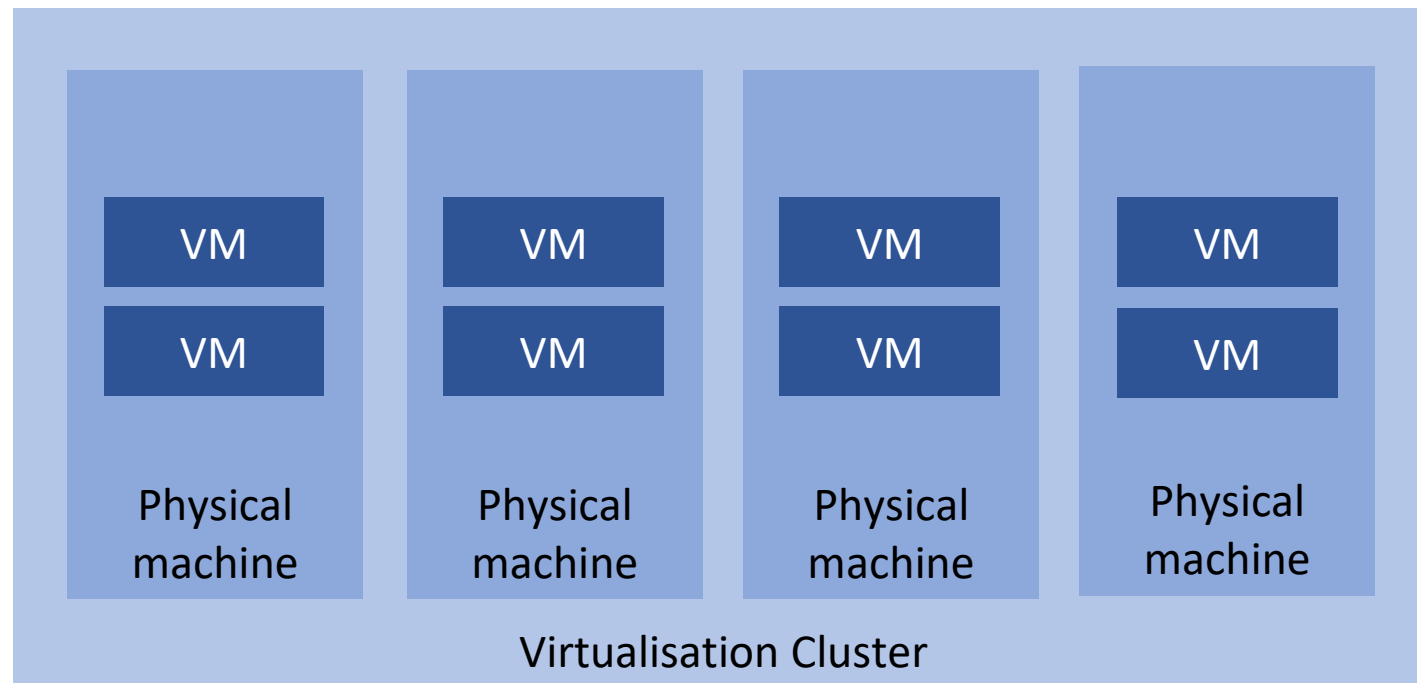
Virtualization availability

- When a physical machine fails, the virtual machines running on it are automatically restarted on the spare physical machine



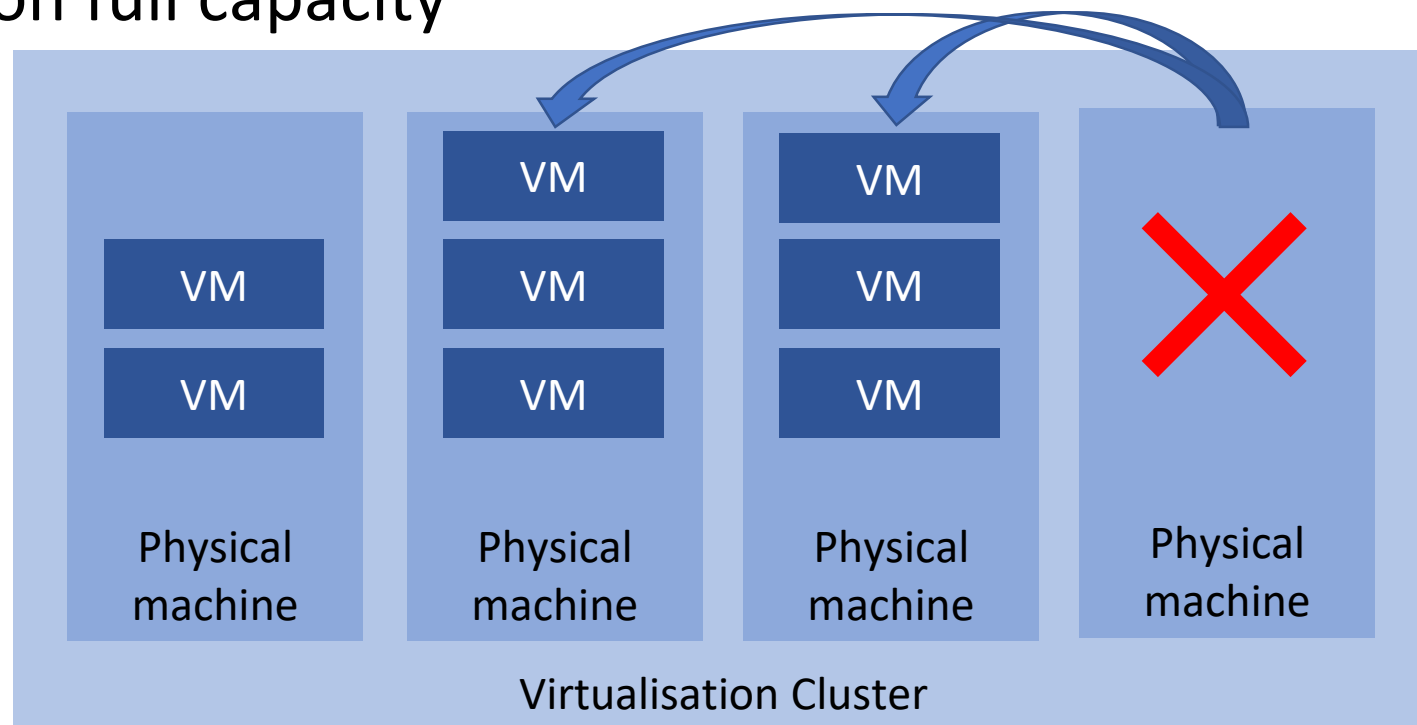
Virtualization availability

- An alternative is to have all physical machines running at lower capacity



Virtualization availability

- When a physical machine fails, the virtual machines running on it are automatically restarted on the other physical machine. All machines now run on full capacity



Compute performance

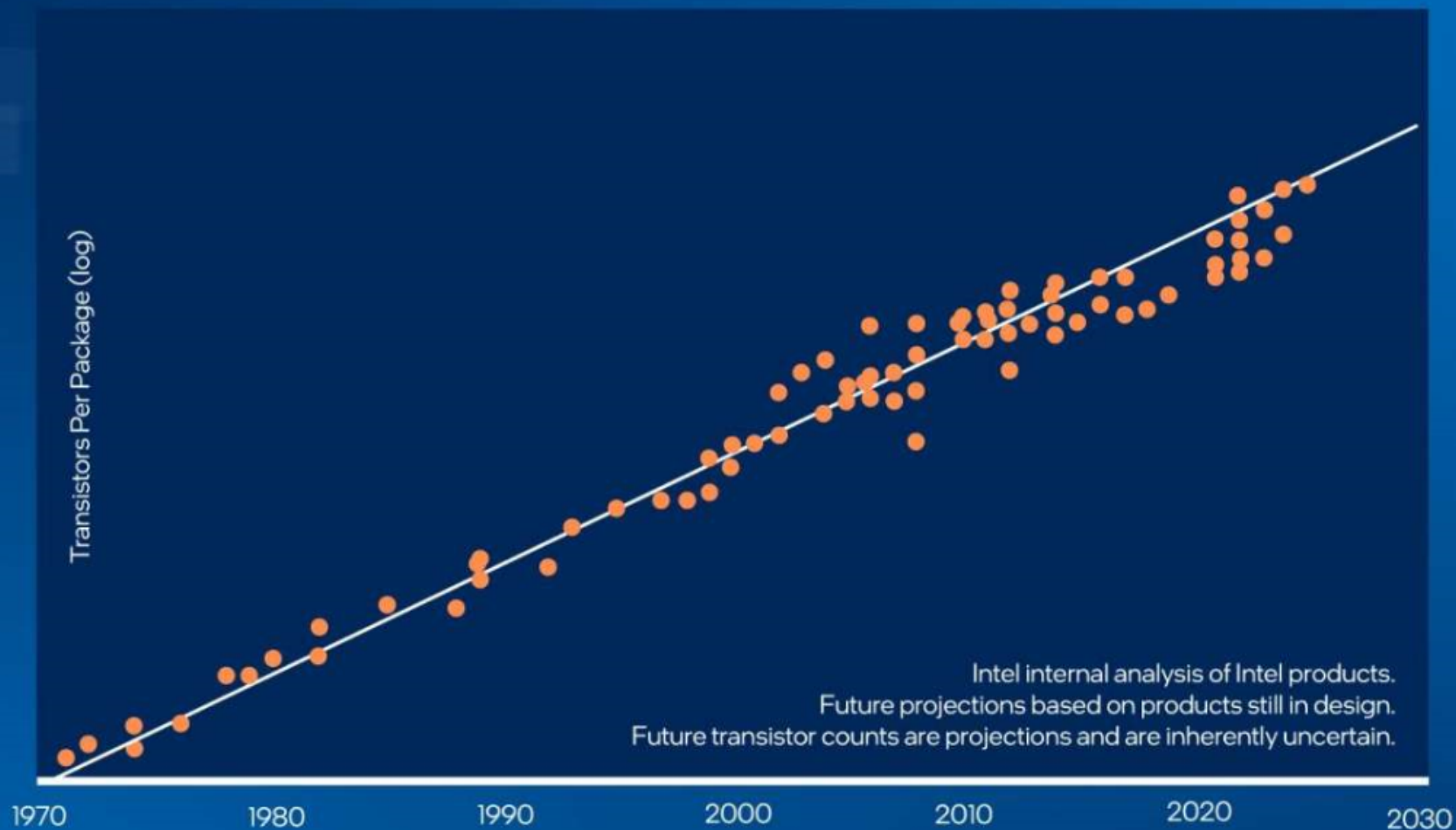
CPU: Moore's law

- In 1971, Intel released the world's first universal microprocessor, the 4004
 - Contained 2,300 transistors
 - Could perform 60,000 instructions per second
 - About as much as the ENIAC computer that filled a complete room and weighed several tons
- Since the introduction of the first CPU in 1971, the power of CPUs has increased exponentially



CPU: Moore's law

- Moore's law states:
 - The number of transistors that can be placed inexpensively on an integrated circuit doubles approximately every two years
- This trend has continued for more than half a century now
- An Intel Alder Lake hybrid processor contains 100,000,000,000 (100 billion) transistors
- A 43 million-fold increase in 52 years' time!

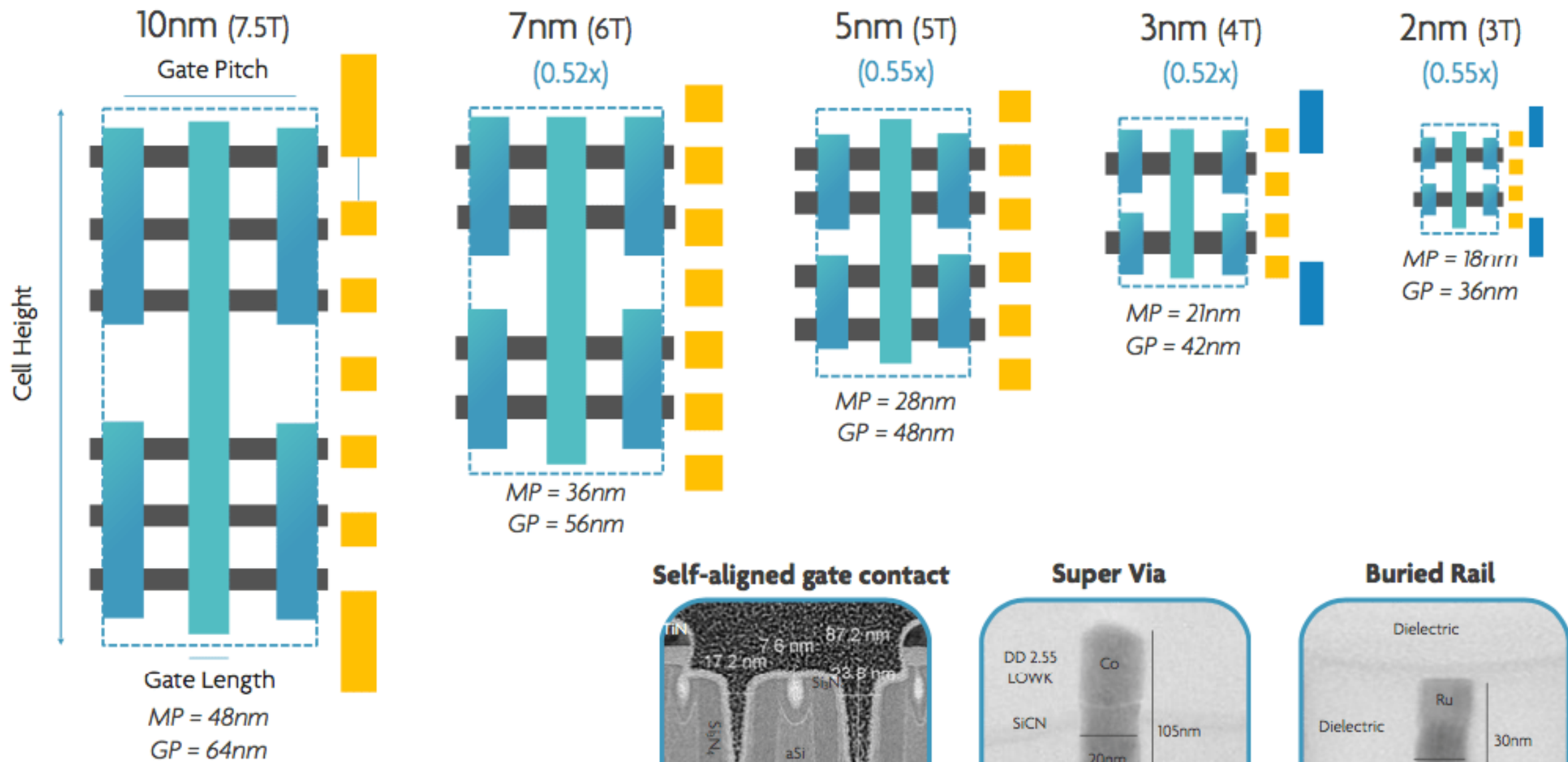


Aspiring to
1 Trillion
transistors in 2030

- ✓ RibbonFET
- ✓ PowerVia
- ✓ High NA
- ✓ 2.5D/3D packaging

CPU: Moore's law

- Moore's law only speaks of the number of transistors; not the performance of the CPU
- The performance of a CPU is dependent on a number of variables, like clock speed, caches and pipelines, data bus width
- Moore's law cannot continue forever, as there are physical limits to the number of transistors a single chip can hold
 - Today, the connections used inside a high-end CPU have a physical width of 5 nm (nanometer)
 - This is extremely small – the size of 24 atoms (the diameter of an atom is of the order of 0.21 nm)



**Cell library scaling
enabled by scaling boosters**



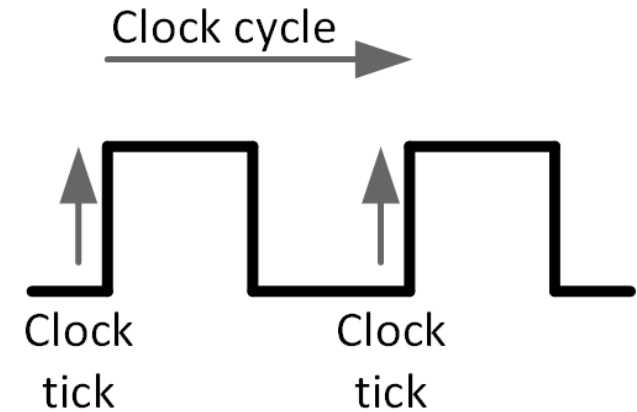
The logo for the Apple M3 chip, featuring the Apple logo and the text "M3" in a white, sans-serif font. The logo is centered within a square frame that has a glowing cyan and green gradient border.The logo for the Apple M3 Pro chip, featuring the Apple logo and the text "M3 PRO" in a white, sans-serif font. The logo is centered within a square frame that has a glowing blue gradient border.The logo for the Apple M3 Max chip, featuring the Apple logo and the text "M3 MAX" in a white, sans-serif font. The logo is centered within a square frame that has a glowing purple gradient border.

CPU: Increasing CPU and memory performance

- Various techniques have been invented to increase CPU performance, like:
 - Increasing the clock speed
 - Caching
 - Prefetching
 - Branch prediction
 - Pipelines
 - Use of multiple cores

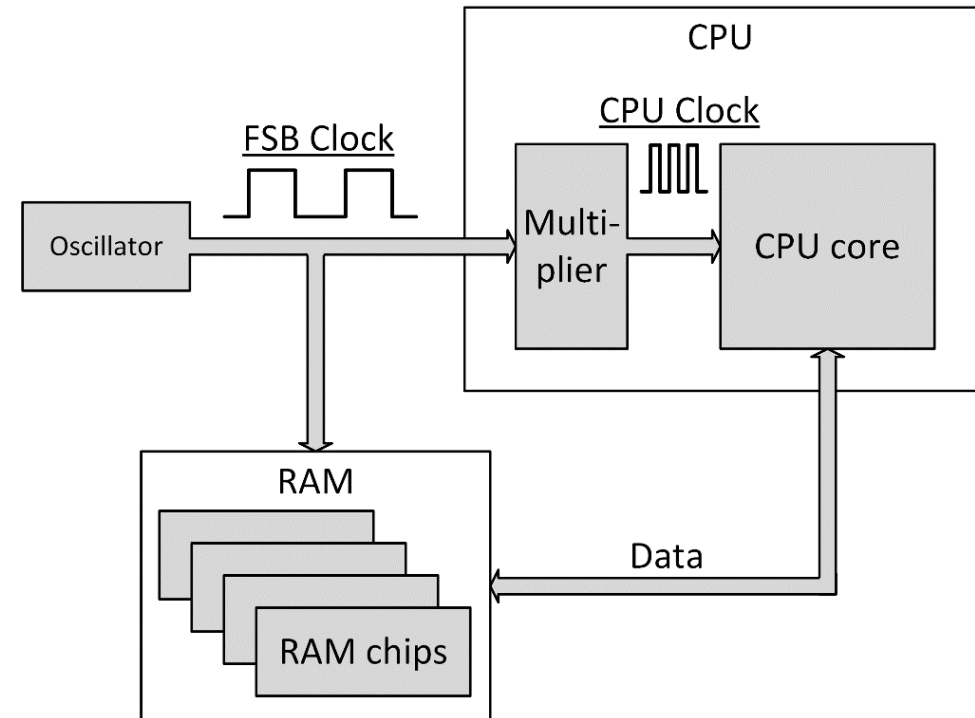
CPU: Increasing clock speed

- CPU instructions are fetched, decoded, executed and stored to memory
- Each step in the sequence is equals to 1 clock tick
- CPU clock speed is measured in Hertz (Hz) – clock ticks or cycles per second



CPU: Increasing clock speed

- Today's CPUs use clock speeds as high as 6 GHz (6,000,000,000 clock ticks per second)
- Because of physical limitations, oscillators cannot run at this speed
- An oscillator with a lower frequency is used (for instance 400 MHz) and this clock rate is multiplied on the CPU chip
- The oscillator speed is known as the Front Side Bus (FSB) speed



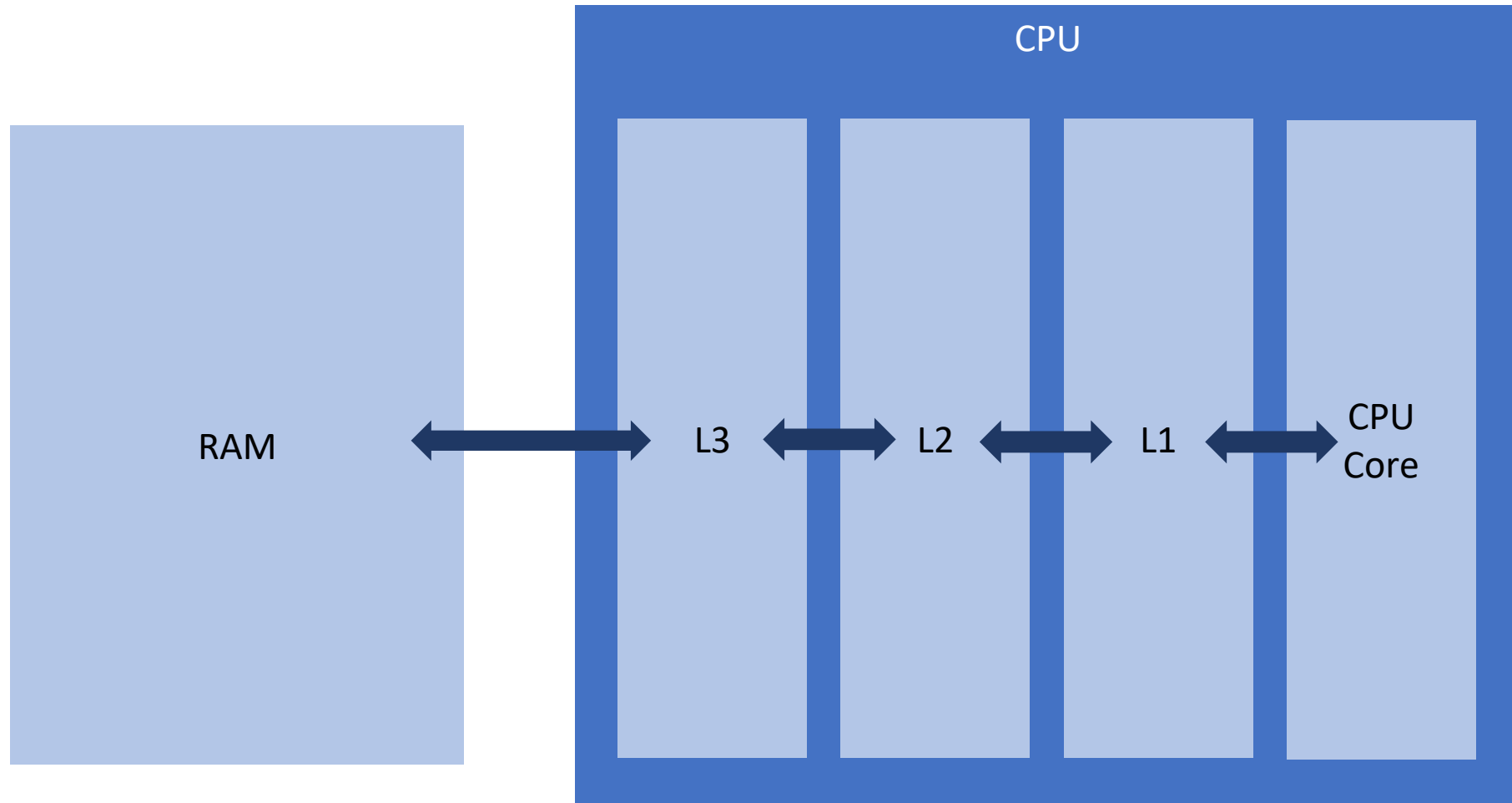
CPU: Caching

- All CPUs in use today contain on-chip caches
- A cache is a relatively small piece of high speed static RAM on the CPU
 - Temporarily stores data received from slower main memory
 - Most CPUs contain two types of cache: level 1 and level 2 cache
 - Some multi-core CPUs also have a large level 3 cache; a cache shared by the cores
- Cache memory runs at full CPU speed (say 3 GHz), main memory runs at the CPU external clock speed (say 100 MHz, which is 30 times slower)

Intel® Core Desktop Processors Comparison

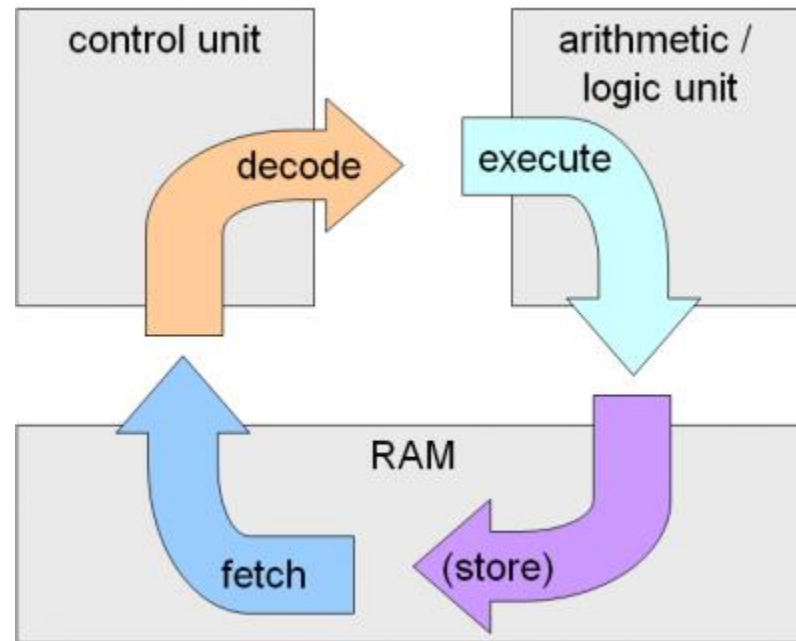
	Intel Core i9 Processors	Intel Core i7 Processors	Intel Core i5 Processors	Intel Core i3 Processors
Max Turbo Frequency [GHz]	Up to 6.0	Up to 5.6	Up to 5.3	Up to 4.7
Intel® Turbo Boost Max Technology 3.0 Frequency [GHz]	Up to 5.8	Up to 5.6	N/A	N/A
Performance-core Max Turbo Frequency [GHz]	Up to 5.6	Up to 5.5	Up to 5.1	Up to 4.7
Efficient-core Max Turbo Frequency [GHz]	Up to 4.4	Up to 4.3	Up to 3.9	N/A
Processor Cores (P-cores + E-cores)	24 (8P + 16E)	20 (8P + 12E)	14 (6P + 8E)	4 (4P + 0E)
Intel® Hyper-Threading Technology	Yes			
Total Processor Threads	32	28	20	8
Intel® Thread Director	Yes			No
Intel® Smart Cache (L3) Size [MB]	36 MB	33 MB	24 MB	12 MB
Total L2 Cache Size [MB]	32 MB	28 MB	20 MB	5 MB

CPU: Caching



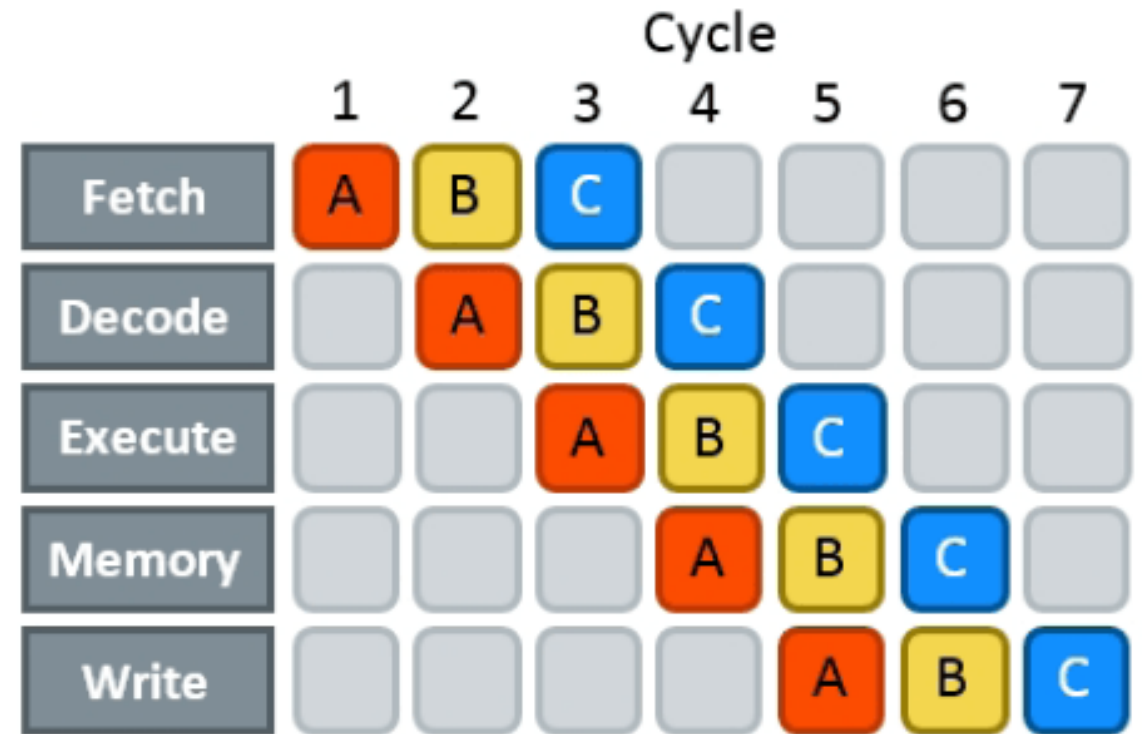
CPU: Pipelines

- Early processors first fetched an instruction, decoded it, then executed the fetched instruction, and wrote the result back before fetching the next instruction and starting the process over again



CPU: Pipelines

- Later CPUs used pipelines
- While the first instruction is being executed, the second instruction can be fetched (since that circuitry is idling anyway), creating instruction overlap



CPU: Prefetching and branch prediction

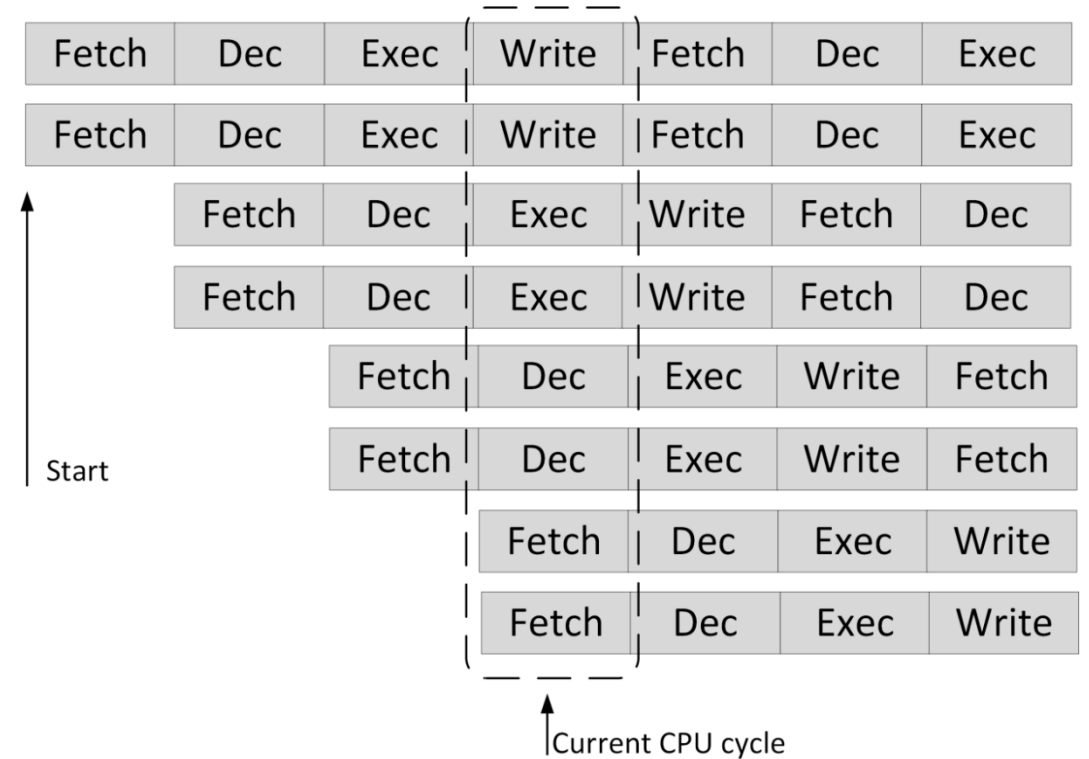
- Prefetching:
 - When an instruction is fetched from main memory, also the next instructions are fetched and stored in cache
 - When the CPU needs the next instruction it is already available in cache
- Unfortunately, most programs contain jumps (also known as branches), resulting in cache misses - the next instruction is not the next instruction in memory

CPU: Prefetching and branch prediction

- The cache system tries to predict the outcome of branch instructions before they are executed by the CPU (called branch prediction)
- In practice more than 80% of the processor instructions are delivered to the CPU from cache memory using prefetching and branch prediction

CPU: Superscalar CPUs

- A superscalar CPU can process more than one instruction per clock tick
- This is done by simultaneously dispatching multiple instructions to redundant functional units on the processor



CPU: Multi-core CPUs

- The fastest commercial CPUs have been between running between 3 GHz and 4 GHz for a number of years now
- Reasons:
 - High clock speeds make connections on the circuit board work as a radio antenna
 - A frequency of 3 GHz means a wavelength of 10 cm. When signals travel for more than a few cm on a circuit board, the signal gets out of phase with the clock
 - The CPU can heat up tremendously at certain spots, which could lead to a meltdown

CPU: Multi-core CPUs

- A multi-core processor is a CPU with multiple separate cores
 - The equivalent of getting multiple processors in one package
- The cores in a multi-core CPU run at a lower frequency
 - Reduce power consumption
 - Reduce heat (no hot spots)
- Trend is to have CPUs with tens or even hundreds of cores

Browse Intel® Xeon® 6 Processors

7 Products [COMPARE ALL](#)

Product Name	Total Cores	Max Turbo Frequency	Processor Base Frequency	Cache
☐ Intel® Xeon® 6740E Processor (96M Cache, 2.40 GHz)	96	3.2 GHz	2.4 GHz	96 MB
☐ Intel® Xeon® 6756E Processor (96M Cache, 1.80 GHz)	128	2.6 GHz	1.8 GHz	96 MB
☐ Intel® Xeon® 6780E Processor (108M Cache, 2.20 GHz)	144	3 GHz	2.2 GHz	108 MB
☐ Intel® Xeon® 6710E Processor (96M Cache, 2.40 GHz)	64	3.2 GHz	2.4 GHz	96 MB
☐ Intel® Xeon® 6766E Processor (108M Cache, 1.90 GHz)	144	2.7 GHz	1.9 GHz	108 MB
☐ Intel® Xeon® 6731E Processor (96M Cache, 2.20 GHz)	96	3.1 GHz	2.2 GHz	96 MB
☐ Intel® Xeon® 6746E Processor (96M Cache, 2.00 GHz)	112	2.7 GHz	2 GHz	96 MB

CPU: Hyper-threading

- Certain Intel CPUs contain a propriety technology called hyper-threading. For example the Core i3/i5/i7 and Xeon CPUs
- Hyper-threading makes a single processor core virtually work as a multi-core processor
- Hyper-threading can provide some increase in system performance by keeping the processor pipelines busier

Virtualization performance

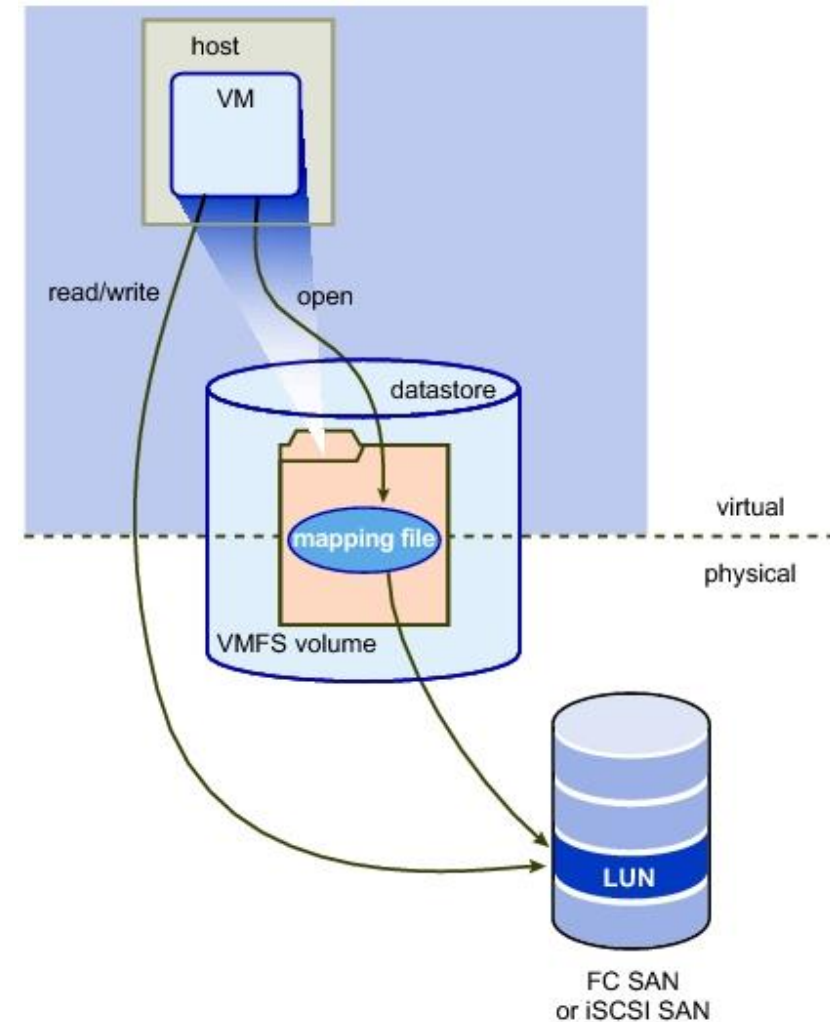
- Consolidating multiple virtual machines on one physical machine increases CPU usage and reduces CPU idle time
- The **physical machine** needs to **handle the disk and network I/O** of all running virtual machines. This can easily lead to an **I/O performance bottleneck**

Virtualization performance

- When choosing a physical machine to host virtual machines, consider getting a machine with:
 - Much **CPU** and **memory** capacity
 - Capability of very **high I/O throughput**
- Virtualization introduces performance penalties:
 - Resources required to run the hypervisor
 - Operation transformations
- This is usually less than 10% of the total performance

Virtualization performance

- **Databases** generally require a **lot of network bandwidth** and **high disk I/O performance**
 - This makes databases less suitable for a virtualized environment
- **Raw Device Mapping** allows a virtual machine exclusive access to a physical storage medium. This diminishes the performance hit of the hypervisor on storage to almost zero



Virtualization performance

- Often **one physical server** is needed **per database**
- Many benefits of virtualization remain
 - Database servers can easily be migrated to other physical machines without downtime
 - Management of the servers is unified when all servers run hypervisors

Compute security

Physical security

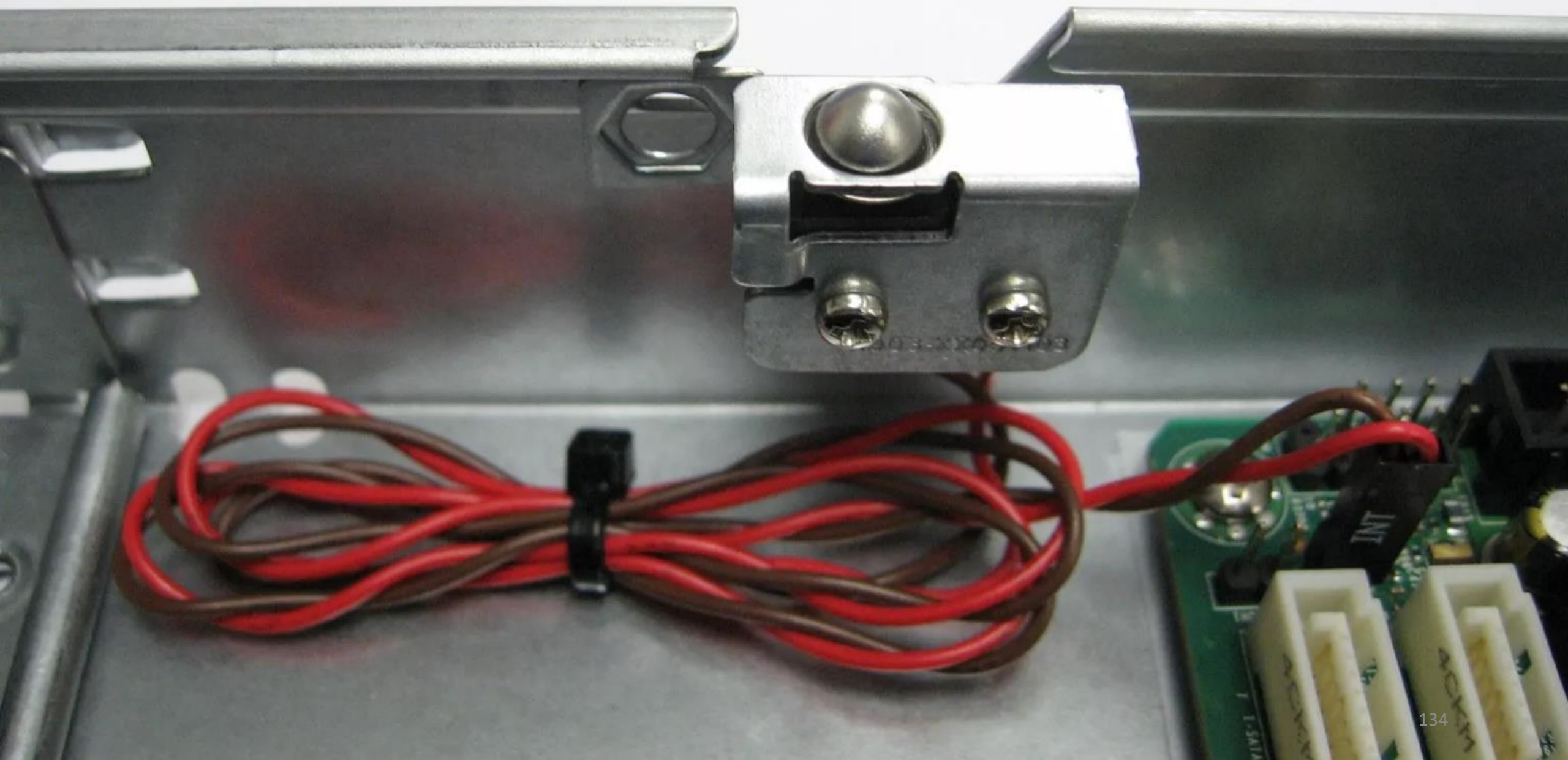
- Disable external USB ports in the BIOS
- BIOS settings in an x86 server should be protected using a password,
 - Via the BIOS, external USB ports can be enabled, and other parameters can be set
- Some servers allow the detection of the physical opening of the server housing
 - Such an event can be sent to a central management console using for instance SNMP traps
 - If possible, enable this to detect unusual activities



Rack Door Contact Sensor

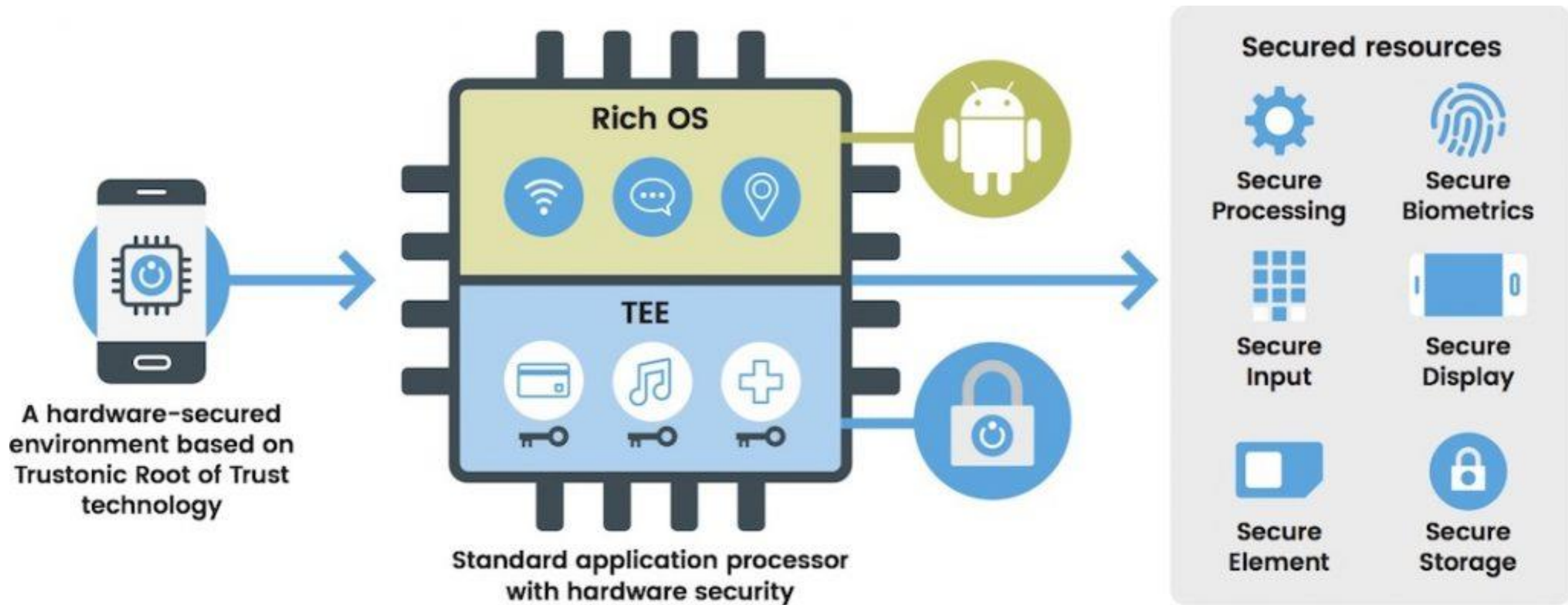


Inductive Proximity Door Sensor



Data in use

- Data in use is data that is actively processed, accessed or used by a system, application or user
- Data in use is an important data privacy and security concern
- Data in use can be protected by implementing a hardware-based Trusted Execution Environment (TEE), which is a secure area of a CPU



Virtualization security

- Virtual machines must be protected the same way as physical machines
- The use of virtualization introduces new security vulnerabilities of its own:
 - If possible, **firewalls** and Intrusion Detection Systems (IDSs) in the hypervisor should be deployed
 - The **virtualization platform** itself needs **patching** too
 - The **size** and **complexity** of the hypervisor should be kept to a **minimum**

Virtualization security

- De-militarized Zone (DMZ)
 - Consider using separate physical machines that run all the virtual machines needed in the DMZ
 - This can improve security, performance, and control
 - It increases costs and complexity

On-premises network
192.168.0.0/16

Azure Vnet
10.0.0.0/16

Gateway subnet
10.0.255.224/27

Public DMZ in
10.0.0.64/27

Web tier
10.0.1.0/24

Gateway

UDR

NVA

NSG

UDR

Availability
set

VM

VM

VM

NSG

PIP

Internet clients

Requests to Azure workload

Requests from Azure workload

Virtualization security

- Systems management console
 - The systems management console connects to all hypervisors and virtual machines
 - If the systems management console is hacked, all virtual machines are affected
 - Not all systems managers should have access to all virtual machines
 - Special user accounts and passwords should be configured for high risk operations like shutting down physical machines or virtualized clusters
 - All user activity in the systems management console should be logged

Conclusion

- Discuss compute building blocks
- Discuss compute availability
- Discuss compute performance
- Discuss computer security